

Transient High-Coherence Neural States with Golden Ratio Harmonic Organization at Earth-Resonant Frequencies

Discovery of Schumann Ignition Events in Human EEG

Michael Lacy

michael.lacy@gmail.com

January 12, 2026

We report the discovery and characterization of Schumann Ignition Events (SIEs)—transient, high-coherence neural states occurring at frequencies within the range of Earth’s Schumann Resonance. Analysis of 1,366 events across 91 participants, 661 recording sessions, and three consumer EEG devices (Emotiv EPOC X, Muse, Insight) spanning five cognitive contexts reveals brief episodes of multi-band network synchronization with consistent spectral architecture.

SIEs are characterized by elevated power and phase coherence across multiple frequency bands simultaneously. The fundamental frequency (7.63 ± 0.33 Hz) falls within the natural fluctuation range of the first Schumann Resonance mode (7.0–8.2 Hz). At this frequency, 81.5% of events exhibited phase-locking values exceeding 0.6, and 60.5% showed magnitude-squared coherence above 0.5. Events followed a stereotyped six-phase temporal evolution in which coherence elevation preceded amplitude increases—a signature consistent with network-level coordination rather than local oscillatory bursts. Detected frequencies converged across devices and cognitive contexts for lower harmonics (SR1, SR3), supporting measurement validity, though higher harmonics showed modest device-related variability.

Harmonic frequency analysis revealed that ratios between detected harmonics approximate powers of the golden ratio ($\phi = 1.618$) with less than 1% error. Critically, individual harmonic frequencies varied independently across events (all pairwise $|r| < 0.04$), yet ratio precision was preserved—indicating that golden ratio relationships are embedded in population-level frequency distributions rather than arising from within-event coupling. Null controls confirmed genuine phase synchronization (phase-randomized surrogates: $d = 2.20$) and spectral exceptionality versus random triplets from actual EEG peaks ($p < 0.0001$, $d = 1.44$). Baseline periods outside detected events showed identical frequency ratios with reduced power and coherence, establishing that SIEs represent transient amplification of continuously present oscillatory structure.

These findings establish SIEs as a reproducible multi-band synchronization phenomenon with precise frequency organization corresponding to the Schumann Resonance range. Whether the SR-neural frequency correspondence reflects functional coupling, convergent biophysical constraints, or coincidence cannot be determined without concurrent geomagnetic measurement. The golden ratio harmonic architecture—present continuously but amplified during SIEs—suggests intrinsic organizational principles in neural oscillatory dynamics warranting further investigation.

Contents

1	Introduction	5
2	Methods	6
2.1	Overview	6
2.2	Ethics Statement	7
2.2.1	Discovery Dataset Ethics	7
2.2.2	AI Tool Disclosure	7
2.3	Participants and Recordings	7
2.3.1	Discovery Dataset: Longitudinal Meditation Recordings	7
2.3.2	Validation Dataset 1: PhySF Cognitive Flow Study	8
2.3.3	Validation Dataset 2: MultiPENG Video Game Engagement Study	8
2.3.4	Validation Dataset 3: Visual Stimuli EEG Study	8
2.3.5	Combined Dataset Summary	8
2.3.6	Sample Size Considerations	9
2.4	EEG Data Acquisition and Preprocessing	9
2.5	Schumann Resonance Harmonic Detection	9
2.5.1	Canonical Frequencies	9
2.5.2	Spectral Parameterization Using FOEOF	10
2.5.3	Hybrid Two-Phase Harmonic Detection	10
2.5.4	Peak Matching Strategy	11
2.6	Ignition Event Detection	11
2.6.1	Schumann Resonance Envelope Computation	12
2.6.2	Onset Detection	12
2.6.3	Window Definition and Merging	12
2.6.4	Neural Ignition Onset (t_0) Refinement	13
2.7	Spatio-Spectral Decomposition	14
2.8	Per-Event Multi-Scale Characterization	15
2.8.1	Per-Harmonic SR Envelope Z-Scores	15
2.8.2	Per-Event Magnitude Squared Coherence (MSC)	15
2.8.3	Per-Event Phase Locking Value (PLV)	16
2.9	Event Quality Metrics	17
2.9.1	Harmonic Stack Index (HSI)	17
2.9.2	Spectral Slope	17
2.9.3	Frequency Specificity Index (FSI)	18
2.10	Event Quality Scoring	18
2.11	Pipeline Summary	19
2.12	Advanced SIE Characterization Methods	19
2.12.1	Time-Frequency Analysis	19
2.12.2	Harmonic Piano Roll Visualization	20
2.12.3	Envelopes and Phase Locking Value (PLV)	21
2.12.4	Harmonic Stack Index (HSI)	21

2.12.5	Triadic Bicoherence (Nonlinear Phase Coupling)	22
2.12.6	Theta-Gamma Phase-Amplitude Coupling (PAC)	23
2.12.7	Six-Phase Evolution Detection Algorithm	24
2.13	Network Dynamics and Connectivity Analysis	25
2.13.1	Harmonic Envelopes	25
2.13.2	Frequency-Specific EEG-SR Coherence	25
2.13.3	Kuramoto Global Phase Synchronization Dynamics	26
2.13.4	Power-Coherence Phase Space Analysis	27
2.13.5	Directed Information Flow Topology (Transfer Entropy Proxy)	27
2.13.6	Cross-Scale Information Transfer (Theta→Gamma)	28
2.14	Null Control Analyses	29
2.14.1	Null Control A: Phase Randomization	29
2.14.2	Null Control B: Random Temporal Windows (Baseline Comparison)	29
2.14.3	Null Control C: Random Spectral Peak Triplets (Uniform)	30
2.14.4	Null Control D: Peak-Based Random Triplets	30
2.14.5	Null Control E: Shuffled Bootstrap (Within-Event Coordination)	31
2.14.6	Null Control F: Distributional Null Model	32
2.15	Cross-Validation Analyses	32
2.15.1	Device Independence	32
2.15.2	Context Independence	32
2.15.3	Task Condition Comparison	32
2.16	Golden Ratio Analysis	32
2.16.1	Predicted Frequencies	32
2.16.2	Ratio Precision Analysis	33
2.16.3	Independence-Convergence Analysis	33
2.17	Statistical Analysis	33
2.17.1	General Approach	33
2.17.2	Multiple Comparisons	33
2.17.3	Non-Parametric Alternatives	34
2.17.4	Bootstrap Confidence Intervals	34
2.18	Data and Code Availability	34
2.18.1	Meditation Dataset	34
2.18.2	Analysis Code	34
2.18.3	Derived Data	34
2.19	Declarations	34
2.19.1	Conflicts of Interest	34
2.19.2	Funding	34
2.20	Software and Computing Environment	34
3	Results	35
3.1	Dataset Overview and Event Detection	35
3.2	Harmonic Definitions	35
3.3	Exemplar Event Walkthrough	36

3.3.1	Spectral and Temporal Dynamics	37
3.3.2	Cross-Frequency Coupling	39
3.3.3	Network Synchronization	40
3.3.4	Information Flow Topology	41
3.3.5	Exemplar Event Summary	42
3.4	Statistical Validation	42
3.4.1	Cross-Device Validation	42
3.4.2	Context Independence	43
3.4.3	Null Control Validation	43
3.4.4	Subject-Level Reproducibility	45
3.5	Synchronization Metrics	45
3.5.1	Power Elevation	46
3.5.2	Coherence and Phase-Locking	46
3.5.3	Cross-Frequency Coupling	47
3.6	Event Quality Assessment	47
3.6.1	SR-Score Distribution	47
3.6.2	Harmonic Stack Index (HSI)	47
3.6.3	Relationship Between Quality and ϕ -Precision	47
3.6.4	Quality Determinants	48
3.7	Temporal Dynamics and Network Organization	48
3.7.1	Six-Phase Temporal Evolution	48
3.8	Emergent Frequency Properties	49
3.8.1	Golden Ratio (ϕ^n) Frequency Organization	49
3.8.2	The Independence-Convergence Paradox	50
3.8.3	Frequency Stability Hierarchy	52
3.8.4	Beat Frequency Architecture	52
4	Discussion	52
4.1	Summary of Principal Findings	52
4.2	Relationship to Transient Synchronization Phenomena	53
4.3	The Independence-Convergence Paradox: A Novel Organizational Principle	54
4.4	Golden Ratio Organization and EEG Frequency Band Structure	54
4.5	Two Synchronization Subsystems	55
4.6	Relationship to Schumann Resonance	56
4.7	Functional Significance of SIEs	57
4.8	Methodological Considerations	58
4.9	Limitations	59
4.10	Future Directions	59
4.11	Conclusions	60

1 Introduction

Neural oscillations are fundamental to brain function, coordinating activity across distributed networks to support perception, cognition, and behavior (Buzsáki and Draguhn, 2004b; Fries, 2015b). These rhythmic fluctuations in neuronal population activity occur across characteristic frequency bands—from slow delta oscillations (<4 Hz) through high gamma (>80 Hz)—each associated with distinct functional roles (Klimesch, 1999b; Engel and Fries, 2010; Cohen, 2017). Although substantial research has characterized sustained oscillatory states and their behavioral correlates, transient synchronization phenomena—brief episodes of enhanced coordination across brain regions—have received comparatively less systematic investigation despite their potential significance for understanding rapid neural state transitions and information integration (Siegel et al., 2012; Canolty and Knight, 2010b).

Well-characterized examples of transient oscillatory events include sleep spindles (Fernandez and Lüthi, 2020), hippocampal sharp-wave ripples (Buzsáki, 2015), and task-related gamma bursts (Fries et al., 2001). These phenomena share common features: discrete temporal boundaries, elevated power relative to baseline, and enhanced inter-regional synchronization. Such transient events may carry functional significance distinct from sustained oscillations, potentially serving as temporal windows for memory consolidation, inter-regional communication, or state transitions (J. Lisman and Jensen, 2013).

The Earth’s ionospheric cavity supports a set of global electromagnetic resonances, termed Schumann Resonances (SR), arising from electromagnetic waves circling the Earth-ionosphere waveguide driven primarily by global lightning activity (Schumann, 1952; Nickolaenko and Hayakawa, 2014). The fundamental SR mode occurs at approximately 7.83 Hz, with higher harmonics at approximately 14.3, 20.8, 27.3, and 33.8 Hz—though these values fluctuate daily between approximately 7.0 and 8.2 Hz for the fundamental, depending on ionospheric conditions and global lightning distribution (Price, 2016). The overlap between SR frequencies and canonical EEG bands—particularly theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz)—has prompted investigation into potential brain-environment relationships, beginning with König’s early observations of spectral similarities between SR and human EEG (König et al., 1981) and continuing through more recent quantitative analyses (Cherry, 2002; Persinger, 2008; Saroka et al., 2016).

Several studies have reported correlations between geomagnetic activity at SR frequencies and various physiological and behavioral measures, including heart rate variability, reaction times, and blood pressure dynamics (Pobachenko et al., 2006; Babayev and Allahverdiyeva, 2007). However, despite this foundational work, critical methodological gaps limit our understanding of SR-aligned neural phenomena. First, prior studies have mainly employed correlational designs examining continuous SR-EEG relationships rather than systematically detecting and characterizing discrete events; the temporal structure, duration, and network dynamics of individual SR-frequency episodes remain poorly characterized. Second, methodological concerns persist: many studies used single EEG devices without cross-platform validation, few employed rigorous null controls to distinguish genuine synchronization from spectral coincidence, and the potential circularity of searching for predefined SR frequencies has not been adequately addressed. Third, harmonic relationships between SR-aligned neural frequencies have received

limited quantitative analysis—particularly whether observed frequency ratios reflect electromagnetic cavity physics, alternative mathematical constraints, or measurement variability.

The present study addresses these gaps through systematic detection and characterization of transient high-coherence neural events occurring at frequencies within the Schumann Resonance range. During exploratory analysis of EEG recordings from meditation sessions collected between 2019 and 2024, we repeatedly observed transient periods of elevated power spectral density at approximately 7.6–7.8 Hz across multiple brain regions. The correspondence with the SR frequency range prompted systematic investigation. We discovered that these events involve coordinated power increases at multiple frequencies, accompanied by marked elevation in inter-channel coherence and phase synchronization—suggesting network-wide coordination rather than isolated oscillatory bursts.

We developed a multi-stage detection pipeline and validated the phenomenon across two independent datasets: longitudinal meditation recordings and a publicly available cognitive task dataset (Albuquerque et al., 2024). To establish event validity, we implemented multiple null control analyses including phase-randomized surrogates, random temporal window sampling, random frequency triplet comparisons, and shuffled bootstrap tests.

Our analyses reveal 1,366 discrete events—which we term Schumann Ignition Events (SIEs)—exhibiting consistent spectral, temporal, and network characteristics across 91 subjects, three EEG devices, and five cognitive contexts. Events show robust network synchronization (phase-locking values exceeding 0.6 in 81.5% of events) with coherence elevation preceding power amplification—a temporal signature suggesting network-level coordination. Harmonic frequency ratios approximate powers of the golden ratio ($\varphi = 1.618$) with less than 1% error, a precision validated as spectrally exceptional compared to random triplets sampled from actual EEG spectral peaks (Cohen’s $d = 1.44$, $p < 0.0001$). Critically, null control analysis revealed that SR-frequency organization exists continuously at baseline with maintained frequency ratios but degraded power and coherence; SIEs represent transient amplification of this persistent oscillatory substrate rather than de novo generation of SR-aligned activity.

These findings establish SIEs as a reproducible neural phenomenon with consistent frequency organization across subjects and recording conditions. The frequencies detected fall within the natural fluctuation range of Earth’s Schumann Resonances (7.0–8.2 Hz), though whether this correspondence reflects functional brain-environment coupling, convergent biophysical constraints, or coincidence cannot be determined without concurrent geomagnetic measurement. We discuss implications for understanding transient synchronization phenomena and the organizational principles underlying neural frequency architecture.

2 Methods

2.1 Overview

This study employed a two-phase design: (1) discovery and initial characterization of Schumann Ignition Events (SIEs) in longitudinal meditation EEG recordings, and (2) validation and extension in an independent dataset collected during cognitive tasks. All analyses were performed in Python 3.9+ using NumPy, SciPy, MNE-Python (Harris et al., 2020; Virtanen et al., 2020;

Gramfort et al., 2013), and the FOOF library (Donoghue et al., 2020). Complete analysis code and parameter specifications will be made available upon publication.

2.2 Ethics Statement

2.2.1 Discovery Dataset Ethics

The longitudinal meditation recordings were collected by the first author from their own personal meditation practice for the purpose of self-directed neurofeedback exploration. As self-experimentation by the researcher on their own data, this component does not constitute human subjects research requiring Institutional Review Board (IRB) approval under 45 CFR 46.102. No identifying information beyond that necessary for scientific reporting is included.

2.2.2 AI Tool Disclosure

In accordance with policy on AI-assisted tools, we disclose the following: AI tools (Claude Opus 4.5, Anthropic) were used for coding assistance in developing the analysis pipeline and for manuscript preparation. Overleaf was used for paper creation. ChatGPT (OpenAI), Google Gemini, and Grok (xAI) were used as manuscript reviewers. All scientific hypotheses, experimental design, data collection, interpretation of results, and intellectual contributions are the authors' own work. AI-generated code was reviewed, tested, and validated by the author prior to use.

2.3 Participants and Recordings

2.3.1 Discovery Dataset: Longitudinal Meditation Recordings

The initial discovery and characterization of SIEs was conducted using EEG recordings from a single experienced meditation practitioner (male, age range 44–48 years, >10 years meditation experience). Recordings were collected during eyes-closed meditation sessions spanning November 2019 to March 2024, comprising 34 sessions with a total recording duration of approximately 7.5 hours. Session durations ranged from 2 to 30 minutes (mean $\pm$ SD: 13.2 $\pm$ 8.7 minutes). Three consumer-grade EEG devices were used across sessions to enable cross-device validation:

- **Muse (2016 model; InteraXon Inc., Toronto, Canada)**- Channels: 4 (TP9, AF7, AF8, TP10; dry electrodes)- Sampling rate: 256 Hz- Reference: FPz- Hardware filters: 0.1–55 Hz bandpass- Sessions recorded: 17
- **Emotiv EPOC X (Emotiv Inc., San Francisco, USA)**- Channels: 14 (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4; saline-soaked felt pads)- Sampling rate: 128 Hz- Reference: CMS/DRL (P3/P4 locations)- Hardware filters: 0.16–43 Hz bandpass, 50/60 Hz notch- Sessions recorded: 5
- **Emotiv Insight (Emotiv Inc.)**- Channels: 5 (AF3, AF4, T7, T8, Pz; semi-dry polymer sensors)- Sampling rate: 128 Hz- Reference: CMS/DRL- Hardware filters: 0.16–43 Hz bandpass, 50/60 Hz notch- Sessions recorded: 12

All recordings were collected in indoor residential environments without electromagnetic shielding. Participants sat in a comfortable upright position with eyes closed. No specific meditation instructions were given beyond maintaining relaxed alertness.

2.3.2 Validation Dataset 1: PhySF Cognitive Flow Study

To test whether SIEs generalize beyond meditation to other cognitive states, we analyzed the publicly available Physiological States of Flow (Albuquerque et al., 2024). This dataset contains multimodal physiological recordings from participants performing mathematical tasks designed to induce flow states (optimal experience characterized by absorbed attention and intrinsic motivation) along with non-flow control conditions.

Original Study Design (Albuquerque et al., 2024)- N = 26 participants; Task: Arithmetic problem-solving; Conditions: Flow (balanced challenge/skill), easy (insufficient challenge), hard (excessive challenge); EEG recording: Emotiv EPOC X (14 channels, 128 Hz); Session duration: 30 minutes per condition

2.3.3 Validation Dataset 2: MultiPENG Video Game Engagement Study

To assess SIE generalization to interactive gaming contexts, we analyzed the publicly available Multimodal Dataset for Player Engagement Analysis in Video Games (Albuquerque et al., 2023).

Original Study Design - N = 39 participants; Task: Playing popular video games across varying difficulty levels; EEG recording: [Device specifications to be confirmed from full paper]; Total sessions: 900 annotated gameplay sessions; Psychological metrics: Engagement, interest, stress, and excitement annotations collected during natural game pauses

2.3.4 Validation Dataset 3: Visual Stimuli EEG Study

To test whether SIEs occur during passive visual processing, we analyzed the publicly available EEG Dataset for Natural Image Recognition through Visual Stimuli (Faisal et al., 2023).

Original Study Design - N = 32 participants (35 initially recruited; 3 excluded via Vividness of Visual Imagery Questionnaire); Task: Passive viewing of natural object images (apple, flower, car, human face); EEG recording: Emotiv EPOC X (14 channels; AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4); Sampling rate: 128 Hz; Electrode system: International 10-20 system; Hardware filters: Standard Emotiv EPOC X bandpass; Software: EMOTIV Pro for data collection and annotation

2.3.5 Combined Dataset Summary

The combined analysis dataset comprised 91 participants across 661 recording sessions, yielding 1,366 detected SIE events. Six independent datasets spanning five cognitive contexts were included: eyes-closed meditation, cognitive flow arithmetic tasks, video game engagement, and passive visual perception. Recordings were collected using three consumer-grade EEG devices—Muse, Emotiv EPOC X, and Emotiv Insight—with the Emotiv EPOC X used across all datasets (Meditation, PhySF, MPENG, VEP), enabling within-device cross-context validation.

Table 1: Dataset Characteristics

Dataset	Participants	Sessions	SIE Events	Device(s)	Context
Meditation (Files)	1	5	38	EPOC X	Eyes-closed meditation
Meditation (Muse)	1	17	165	Muse	Eyes-closed meditation
Meditation (Insight)	1	12	57	Insight	Eyes-closed meditation
PhySF	25	46	447	EPOC X	Cognitive flow/non-flow tasks
MultiPENG	36	533	604	EPOC X	Video game engagement
VEP	29	48	55	EPOC X	Passive visual perception
Combined	91	661	1,366	3 types	5 contexts

2.3.6 Sample Size Considerations

This study used a discovery-validation design rather than a hypothesis-testing framework. The discovery phase analyzed all available longitudinal meditation recordings (34 sessions). For validation datasets, participants were included if at least one SIE was detected in their recordings; participants with zero detected events were excluded from analysis.

Exclusions by dataset: PhySF: 1 of 26 original participants excluded (3.8%); MultiPENG: 3 of 39 excluded (7.7%); VEP: 3 of 32 excluded (9.4%). Combined validation sample: 90 of 97 original participants (92.8%) exhibited at least one SIE.

Post-hoc sensitivity analysis: With 1,366 total events and observed effect sizes ($d = 1.44$ for peak-based random triplet comparison, $p < 0.0001$), statistical power exceeded 0.99 for primary comparisons.

2.4 EEG Data Acquisition and Preprocessing

Continuous EEG was recorded using three consumer-grade devices: (1) Emotiv EPOC X with 14 channels (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4) at 128 Hz using saline-soaked felt sensors and CMS/DRL reference; (2) Emotiv Insight with 5 channels (AF3, AF4, T7, T8, Pz) at 128 Hz using semi-dry polymer sensors and CMS/DRL reference; and (3) Muse (2016 model) with 4 channels (TP9, AF7, AF8, TP10) at 256 Hz using dry electrodes with FPz reference. All electrodes were positioned according to the international 10-20 system (Jasper, 1958). Raw EEG signals were high-pass filtered at 1 Hz using a 4th-order zero-phase Butterworth filter (Butterworth, 1930) to remove slow drifts while preserving oscillatory activity in the Schumann Resonance frequency range. Data were segmented into continuous recording sessions for subsequent analysis.

2.5 Schumann Resonance Harmonic Detection

2.5.1 Canonical Frequencies

The Earth’s Schumann Resonances arise from electromagnetic standing waves in the Earth-ionosphere cavity, with theoretical frequencies at approximately 7.83 Hz (fundamental) and harmonics near 14.3, 20.8, 27.3, and 33.8 Hz (Schumann, 1952; Nickolaenko and Hayakawa, 2014). However, measured SR frequencies vary with ionospheric conditions. We derived canonical frequencies and search bandwidths from long-term magnetometer data recorded by the Space Observing System 70 in Tomsk, Russia (https://sos70.ru/?page_id=48): SR1 = 7.6 Hz (± 0.6

Hz), SR2o = 13.75 Hz (± 0.75 Hz), SR3 = 20 Hz (± 1.0 Hz), SR4 = 25 Hz (± 2.0 Hz), and SR5 = 32 Hz (± 2.5 Hz). These empirically-derived values served as canonical targets for harmonic detection, with session-specific refinement described below.

2.5.2 Spectral Parameterization Using FOOOF

To separate oscillatory (periodic) activity from aperiodic (1/f) background noise, we first estimated power spectra using Welch’s method (Welch, 1967) and then employed spectral parameterization with the FOOOF algorithm (Haller et al., 2018).

$$\log_{10}(PSD(f)) = L(f) + \sum_i G_i(f)$$

where L represents the aperiodic component modeled as:

$$L(f) = b - \chi \times \log_{10}(f)$$

Table 1: Equation 1 Parameters

$PSD(f)$	Power spectral density at frequency f	measured	$\mu V^2/Hz$
$L(f)$	Aperiodic (1/f) component	fitted	$\log_{10}(\mu V^2/Hz)$
$G_i(f)$	i -th Gaussian periodic peak	fitted	$\log_{10}(\mu V^2/Hz)$
f	Frequency	1–50	Hz

Table 2: Equation 2 Parameters

b	Aperiodic offset (y-intercept)	fitted	$\log_{10}(\mu V^2/Hz)$
χ	Aperiodic exponent (spectral slope)	1–2	dimensionless
f	Frequency	1–50	Hz

Table 3: FOOOF Fitting Parameters

Parameter	Symbol	Value	Units	Rationale
Frequency range	$[f_{\min}, f_{\max}]$	(1.0, 50.0)	Hz	Covers all SR harmonics
Segment length	T_{seg}	100.0	s	Provides 0.01 Hz frequency resolution
Overlap	—	0.5	fraction	50% Welch segment overlap
Max peaks	$n_{\max}$	7	count	Sufficient for 5+ SR harmonics
Peak threshold	θ_{peak}	0.1	$\log_{10}$ units	Lowered for subtle SR peaks
Min peak height	$h_{\min}$	0.05	$\log_{10}$ units	Floor for peak detection
Peak width limits	$[\sigma_{\min}, \sigma_{\max}]$	(0.3, 4.0)	Hz	Accommodates narrow SR peaks

2.5.3 Hybrid Two-Phase Harmonic Detection

We implemented a hybrid approach combining session-level stability with event-level adaptability:

- **Phase 1 (Session-Level):** Power spectral density was computed across the entire recording session using Welch’s method with the parameters specified in Table 3. FOOOF was fitted to extract session-level harmonic frequencies and aperiodic parameters (exponent χ , goodness-of-fit R^2).
- **Phase 2 (Per-Event):** For each detected ignition event (Section 2.3), FOOOF was refitted to the event-specific time window using canonical harmonics as seed frequencies. This captured dynamic frequency shifts during ignition states while maintaining comparability across events.

2.5.4 Peak Matching Strategy

When multiple spectral peaks occurred within a harmonic search window, we employed maximum power selection to identify the matched frequency. For each canonical harmonic f_{can} , all FOOOF-detected peaks within the search window $[f_{\text{can}} - \Delta f, f_{\text{can}} + \Delta f]$ were identified, and the peak with highest spectral power was selected:

$$f_{\text{matched}} = f_k \text{ where } k = \arg \max_i (P_i)$$

Symbol	Definition	Value	Units
f_i	Center frequency of i -th candidate peak	detected	Hz
P_i	Power (height) of i -th candidate peak	detected	$\log_{10}(\mu\text{V}^2/\text{Hz})$
<i>Canonical frequencies and search half-bandwidths (from Tomsk SOS70):</i>			
$f_{\text{can},\text{SR1}}$	SR1 canonical frequency	7.6 ± 0.6	Hz
$f_{\text{can},\text{SR2o}}$	SR2o canonical frequency	13.75 ± 0.75	Hz
$f_{\text{can},\text{SR3}}$	SR3 canonical frequency	20 ± 1.0	Hz
$f_{\text{can},\text{SR4}}$	SR4 canonical frequency	25 ± 2.0	Hz
$f_{\text{can},\text{SR5}}$	SR5 canonical frequency	32 ± 2.5	Hz

The search window for each canonical frequency was defined as $[f_{\text{can}} - \Delta f_{\text{search}}, f_{\text{can}} + \Delta f_{\text{search}}]$, with harmonic-specific half-bandwidths (Table 4) reflecting greater natural variability at higher SR modes. This maximum power selection approach ensures that the strongest spectral component within each search window is matched to the corresponding SR harmonic, prioritizing signal strength over proximity to canonical frequency.

2.6 Ignition Event Detection

We developed a seven-stage detection pipeline to systematically identify, refine, and validate candidate SIE events. The pipeline was designed to balance sensitivity (detecting true SIEs) against specificity (excluding artifacts and non-SIE oscillatory events).

2.6.1 Schumann Resonance Envelope Computation

To detect transient increases in Schumann-band activity, we computed a composite SR envelope across all electrodes using the following procedure:

1. EEG signals from all $N = 14$ channels were averaged to create a composite signal, reducing channel-specific noise while preserving spatially coherent activity.
2. The composite signal was bandpass filtered at the fundamental frequency ($f_0 \pm \Delta f$) using a 4th-order Butterworth filter with zero-phase (filtfilt) implementation.
3. The analytic signal was obtained via Hilbert transform (Gabor, 1946), and the instantaneous envelope was computed as the magnitude of the analytic signal.
4. The envelope was smoothed using a Hanning-windowed moving average of duration T_{smooth} .
5. The smoothed envelope was z-score normalized across the entire session:

$$z(t) = [env(t) - \mu_{env}] / \sigma_{env}$$

Symbol	Definition	Value	Units
$env(t)$	Instantaneous Hilbert envelope magnitude	computed	μV
μ_{env}	Session mean of smoothed envelope	computed	μV
σ_{env}	Session standard deviation of envelope	computed	μV
f_0	Center frequency (fundamental)	7.6	Hz
Δf	Half-bandwidth	0.6	Hz
T_{smooth}	Smoothing window duration	0.01	s
N	Number of electrodes	14	count
f_s	Sampling rate	128	Hz

2.6.2 Onset Detection

Candidate ignition onsets were identified as rising-edge threshold crossings where $z(t) \geq z_{\text{thresh}}$.

Parameter	Symbol	Value	Units	Description
Z-threshold	z_{thresh}	3.0	SD	Detection threshold ($p < 0.003$ under Gaussian assumptions)
Min ISI	T_{ISI}	2.0	s	Minimum inter-stimulus interval to prevent rapid false positives
Filter order	n_{order}	4	—	Butterworth filter order

2.6.3 Window Definition and Merging

For each detected onset, an analysis window was defined with the following parameters: Adjacent windows separated by less than T_{merge} were merged into single extended events to capture prolonged ignition states. All windows were clipped to recording boundaries.

Parameter	Symbol	Value	Units	Description
Window duration	T_{win}	20.0	s	Total window length
Pre-onset	T_{pre}	10.0	s	Time before onset (window centered on onset)
Post-onset	T_{post}	10.0	s	Time after onset
Merge gap	T_{merge}	10.0	s	Maximum gap for merging adjacent windows

2.6.4 Neural Ignition Onset (t_0) Refinement

The precise timing of neural ignition onset (t_0) was refined using Kuramoto phase synchronization (Kuramoto, 1984; Strogatz, 2000). The Kuramoto order parameter $R(t)$ quantifies instantaneous phase coherence across channels:

$$R(t) = \left| \frac{1}{N} \sum_k e^{i\phi_k(t)} \right|$$

The ignition onset t_0 was defined as the time of maximum gradient (dR/dt) where $R(t) \geq R_{\text{thresh}}$, marking the steepest rise in multi-channel phase synchronization—the putative moment of neural ignition.

Procedure:

1. Bandpass filter all channels at SR1 center frequency ± 1.5 Hz (using event-specific SR1 from Stage 2)
2. Compute analytic signal via Hilbert transform
3. Extract instantaneous phase $\varphi_j(t)$ for each channel j
4. Compute Kuramoto order parameter: $R(t) = \frac{1}{N} \left| \sum_{j=1}^N e^{i\varphi_j(t)} \right|$, where N is the number of artifact-free channels.
5. Compute temporal derivative dR/dt using centered finite differences
6. Detect synchronization onset (t_0) at the time of maximum dR/dt where $R(t) > 0.6$
7. Record $R(t_0)$, maximum R during event, and synchronization onset latency relative to power onset

Symbol	Definition	Value	Units
$R(t)$	Kuramoto order parameter (phase coherence)	0–1	dimensionless
N	Number of EEG channels	14	count
$\varphi_k(t)$	Instantaneous phase of channel k at SR frequency	computed	radians
i	Imaginary unit	$\sqrt{-1}$	—

Parameter	Symbol	Value	Units	Description
Sliding window	T_R	2.5	s	$R(t)$ computation window
Step size	Δt_R	0.001	s	Temporal resolution (1 ms)
R threshold	R_{thresh}	0.6	dimensionless	Minimum R for t_0 detection
Edge buffer	T_{edge}	2.0	s	Excluded edge samples to avoid boundary artifacts
Min RMS	RMS_{min}	1×10^{-7}	μV	Power threshold for valid R computation

2.7 Spatio-Spectral Decomposition

To construct an optimized virtual Schumann Resonance reference signal, we employed Spatio-Spectral Decomposition (Nikulin et al., 2011), a spatial filtering technique that maximizes signal-to-noise ratio at target frequencies.

$$C_S w = l C_N w$$

Symbol	Definition	Dimensions	Description
C_S	Signal covariance matrix	$N \times N$	Covariance of SR-band filtered EEG
C_N	Noise covariance matrix	$N \times N$	Covariance of flanking-band filtered EEG
$\mathbf{w}$	Spatial filter weights	$N \times 1$	Eigenvector solution
λ	Eigenvalue	scalar	Signal-to-noise ratio
N	Number of channels	14	—

The eigenvector $\mathbf{w}^*$ corresponding to the maximum eigenvalue λ_{max} provides the optimal spatial filter:

$$\mathbf{w}^* = \text{argmax}_{\mathbf{w}} (\mathbf{w}^T C_S \mathbf{w}) / (\mathbf{w}^T C_N \mathbf{w})$$

Band	Frequency Range	Units	Description
Signal band	$[f_0 - \Delta f, f_0 + \Delta f] = [7.33, 8.33]$	Hz	SR fundamental ± 0.5 Hz
Lower flank	$[f_0 - \Delta f - \Delta f_{\text{flank}}, f_0 - \Delta f] = [6.33, 7.33]$	Hz	Below signal band
Upper flank	$[f_0 + \Delta f, f_0 + \Delta f + \Delta f_{\text{flank}}] = [8.33, 9.33]$	Hz	Above signal band

Parameter	Symbol	Value	Units	Description
Signal bandwidth	Δf	0.4–0.5	Hz	Half-width of signal band
Flank width	Δf_{flank}	1.0	Hz	Width of each flanking noise band
Regularization	ε	1×10^{-12}	—	Pseudoinverse stabilization constant

The virtual reference signal was computed as $\mathbf{v}_{\text{SR}}(t) = \mathbf{w}^{*T} \times \mathbf{X}(t)$, where $\mathbf{X}(t)$ is the $N \times T$ multi-channel EEG matrix. This yields a single-channel reference with enhanced sensitivity to spatially coherent Schumann-band activity while suppressing broadband noise.

2.8 Per-Event Multi-Scale Characterization

For each detected ignition event, multiple time-resolved metrics were computed to characterize the dynamics of Schumann-band activity.

2.8.1 Per-Harmonic SR Envelope Z-Scores

For each harmonic frequency f_n , the envelope z-score was computed within the event window using the same procedure as Section 2.3.1.

Metric	Symbol	Window	Description
Peak z-score	$sr\{n\}_{z_max}$	full event	Maximum $z(t)$ in event window
Mean ± 5 s	$sr_z_mean_pm5$	$t_0 \pm 5$ s	Mean z-score around ignition onset
Post-peak mean	$sr_z_mean_post5$	peak $\rightarrow +5$ s	Decay assessment after peak
Peak time	$sr_z_peak_t$	full event	Time of maximum z-score

These metrics were computed independently for each harmonic ($sr1_z_max$, $sr2_z_max$, $sr3_z_max$, etc.).

2.8.2 Per-Event Magnitude Squared Coherence (MSC)

Mean squared coherence (Carter et al., 1973) was computed for each harmonic frequency using a Hilbert-based approach with median reference:

$$MSC = \frac{|S_{xy}(f)|^2}{S_{xx}(f) \cdot S_{yy}(f)}$$

Procedure:

1. Compute the cross-spectral density $S_{xy}(f)$ between each channel and the median reference signal (median voltage across channels at each time point)
2. Compute auto-spectral densities $S_{xx}(f)$ and $S_{yy}(f)$
3. Calculate MSC: $MSC(f) = \frac{|S_{xy}(f)|^2}{S_{xx}(f) \cdot S_{yy}(f)}$
4. Average MSC across channel pairs within each SR harmonic band (± 1 Hz around detected peak)
5. Record mean, maximum, and bandwidth of MSC peak

Symbol	Definition	Value	Units
z_{ch}	Analytic signal of channel (Hilbert transform)	computed	complex
z_{ref}	Analytic signal of median reference across channels	computed	complex
z_{ref}^*	Complex conjugate of reference	computed	complex
$\langle \cdot \rangle$	Time average over analysis window	—	—

Parameter	Symbol	Value	Units	Description
Sliding window	T_{msc}	1.0	s	MSC computation window
Step size	Δt_{msc}	0.1	s	100 ms temporal resolution
Local window	T_{local}	± 2.5	s	Window around t_0 for peak/mean extraction
Baseline window	$T_{\text{base_msc}}$	-5 to -2	s	Pre-event baseline (relative to t_0)
Bandpass half-width	bw_{msc}	0.5	Hz	Per-harmonic filtering

Metric	Symbol	Description
Peak MSC	msc_peak	Maximum MSC in local window
Mean local MSC	msc_mean_loc	Mean MSC in ± 2.5 s around t_0
Baseline MSC	msc_base	Mean MSC in baseline window
MSC AUC	msc_auc_loc	Area under MSC curve in local window

2.8.3 Per-Event Phase Locking Value (PLV)

Phase locking value (Lachaux et al., 1999) was computed for each harmonic frequency within a ± 5 second window centered on the ignition onset t_0 .

$$PLV = \frac{1}{T} \left| \sum_{t=1}^T e^{i\Delta\phi_{xy}(t)} \right|$$

Procedure:

1. Bandpass filter at each SR harmonic frequency (± 1.5 Hz)
2. Extract instantaneous phase via Hilbert transform
3. Compute pairwise phase differences $\Delta\phi_{xy}(t) = \phi_x(t) - \phi_y(t)$
4. Calculate PLV: $PLV = \frac{1}{T} \left| \sum_{t=1}^T e^{i\Delta\phi_{xy}(t)} \right|$, where T is the number of time points in the event window.
5. Average PLV across all channel pairs
6. Record PLV for each SR harmonic

Symbol	Definition	Value	Units
$\Delta\phi(t)$	Phase difference between channel pairs	computed	radians
T	Number of time samples in analysis window	computed	count
PLV	Phase locking value	0–1	dimensionless
T_{PLV}	Analysis window around t_0	± 5	s

Metric	Symbol	Description
Mean PLV	$sr\{n\}_{plv}$	Mean PLV in analysis window

2.9 Event Quality Metrics

Each detected ignition event was characterized by multiple metrics assessing spectral content, cross-frequency relationships, and artifact contamination.

2.9.1 Harmonic Stack Index (HSI)

The HSI quantifies the degree to which overtone (harmonic) power dominates over fundamental power. Higher HSI values indicate overtone-dominated events; lower HSI values indicate fundamental-dominated events.

Procedure:

1. Bandpass filter the signal at the fundamental frequency ($f_0 \pm \Delta f_0$) and compute mean power: $P_f = \langle x_{f_0}^2 \rangle$
2. For each overtone frequency f_n (excluding the fundamental), bandpass filter and compute mean power: $P_n = \langle x_{f_n}^2 \rangle$
3. Sum all overtone powers: $P_{\text{overtones}} = \sum_n P_n$
4. Calculate Harmonic Stack Index:

$$HSI = \frac{P_{\text{overtones}}}{P_f}$$

Symbol	Definition	Value	Units
P_f	Mean power at fundamental frequency	computed	μV^2
$P_{\text{overtones}}$	Sum of mean powers at all overtone frequencies	computed	μV^2
Δf_0	Fundamental band half-width	0.5	Hz
Δf_n	Overtone band half-width	0.5	Hz
$HSI > 1$	Overtone-dominated (harmonics stronger than fundamental)	—	—
$HSI < 1$	Fundamental-dominated (fundamental stronger than harmonics)	—	—

2.9.2 Spectral Slope

The aperiodic exponent β was computed from the Welch periodogram (3–45 Hz, excluding SR frequencies) via linear regression in log-log space.

Parameter	Symbol	Value	Units	Interpretation
Frequency range	$[f_{\beta_min}, f_{\beta_max}]$	[3, 45]	Hz	Broadband range
Typical neural β	β_{neural}	1–2	dimensionless	Physiological activity
Artifact indicator	$\beta_{artifact}$	< 0.5	dimensionless	Flattened slope suggests broadband contamination

2.9.3 Frequency Specificity Index (FSI)

The FSI measures the concentration of power at SR frequencies relative to total broadband power:

$$FSI = P_{SR}/P_{total}$$

Symbol	Definition	Value	Units
P_{SR}	Summed power within $\pm\Delta f$ of each harmonic	computed	μV^2
P_{total}	Total power in [3, 45] Hz	computed	μV^2
Δf_{FSI}	Band half-width around each harmonic	0.5	Hz
FSI > 0.35	Indicates frequency-specific activity (true ignition)	—	fraction
FSI < 0.05	Indicates broadband contamination (artifact)	—	fraction

2.10 Event Quality Scoring

A composite quality score is computed for each ignition event to facilitate ranking and selection, using only the three observed Schumann harmonics (sr1, sr3, sr5):

$$SR\text{-Score} = \frac{Z_{weighted}^{0.7} \times MSC_{weighted}^{1.2} \times PLV_{weighted}}{1 + HSI}$$

Symbol	Definition	Value	Units
$Z_{weighted}$	Weighted z-scores across sr1, sr3, sr5	computed	SD
$MSC_{weighted}$	Weighted coherence across sr1, sr3, sr5	computed	0–1
$PLV_{weighted}$	Weighted PLV across sr1, sr3, sr5	computed	0–1
HSI	Harmonic stack index (penalty for overtone dominance)	computed	ratio
α	Z-score exponent	0.7	—
β	MSC exponent	1.2	—

The weighted averaging sums contributions from the three observed Schumann harmonics using fixed weights derived from the golden ratio:

$$X_{weighted} = \sum_{n \in \{1,3,5\}} w_n \times X_{sr_n}$$

Table 2: Fixed weights for observed Schumann harmonics

Label	Frequency (Hz)	Weight	Approximate φ^{-e}
sr1	~ 7.6	0.618	φ^{-1}
sr3	~ 20	0.326	$\varphi^{-2.5}$
sr5	~ 32	0.146	φ^{-4}

This weighting scheme emphasizes the fundamental frequency (sr1) while progressively attenuating higher harmonic contributions. The HSI penalty in the denominator discourages overtone-dominated events that may reflect non-SR oscillatory activity. By restricting scoring to the three observed Schumann resonance frequencies, this metric enables direct comparison across sessions with different FOOOF-detected harmonic structures.

2.11 Pipeline Summary

Stage	Purpose	Key Parameters	Output
1	Envelope thresholding	$z > 3.0$, duration ≥ 3 s	Candidate windows
2	Harmonic refinement	FOOOF, 3 search windows	Peak frequencies
3	Network synchronization	Kuramoto $R(t)$, $R > 0.6$	Synchronization onset
4	Coherence characterization	MSC, PLV per harmonic	Inter-channel coupling
5	Harmonic organization	HSI computation	Overtone/fundamental ratio
6	Quality metrics	Slopes, durations	Event characterization
7	Composite scoring	SR-Score (sr1, sr3, sr5)	Event quality index

2.12 Advanced SIE Characterization Methods

Beyond the core detection and quality metrics described above, each Schumann Ignition Event (SIE) was further characterized using advanced signal processing techniques to probe the underlying neural dynamics. These methods assess time-frequency structure, nonlinear phase coupling, cross-frequency interactions, and temporal phase evolution that may distinguish genuine brain-field coupling from spurious oscillatory events. The following subsections detail complementary analyses applied to validated SIE windows.

2.12.1 Time-Frequency Analysis

Time-frequency decomposition was performed using Short-Time Fourier Transform (STFT) (Allen, 1977) with the following parameters:

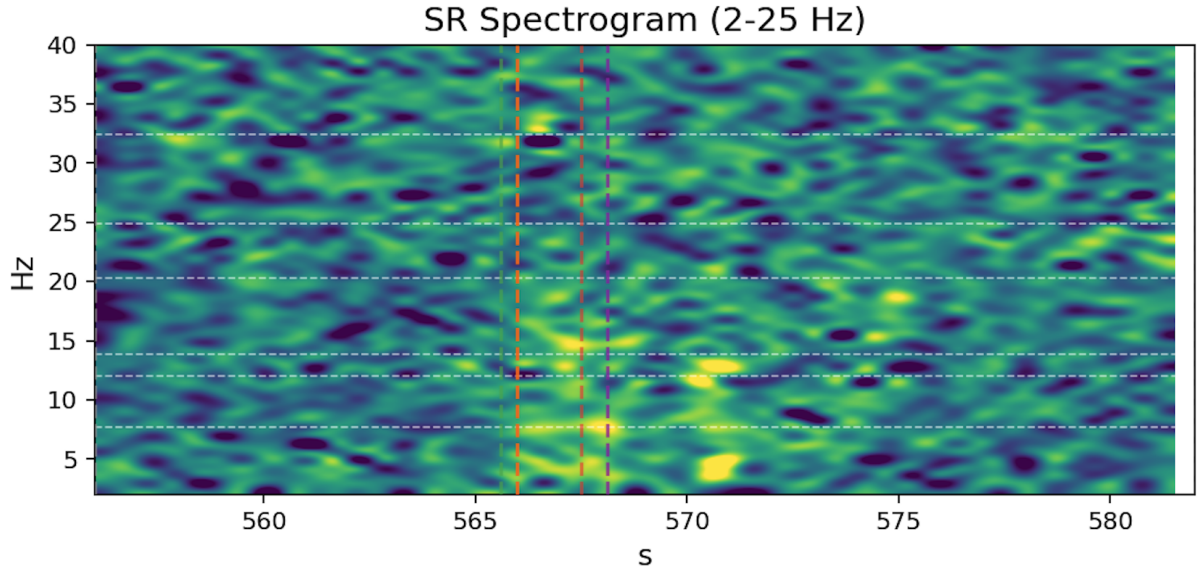


Figure 1: Enter Caption

Parameter	Value	Units	Description
Window duration	2–4	s	Adaptive based on frequency resolution requirements
Overlap	75–95	%	High overlap for smooth temporal dynamics
Frequency range	2–40	Hz	Covers all SR harmonics and gamma
Aggregation	median	—	Robust to single-channel artifacts
Normalization	row-wise z-score	—	Highlights temporal dynamics, controls for 1/f decay

2.12.2 Harmonic Piano Roll Visualization

Envelope z-scores for each detected SR harmonic were displayed as a heatmap representation:

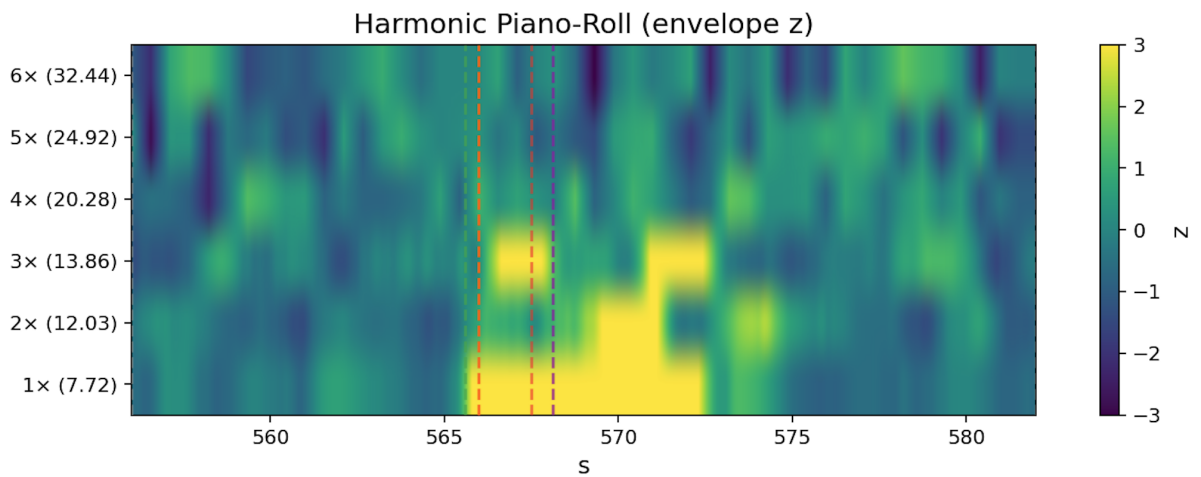


Figure 2: Enter Caption

- **Rows:** Individual harmonics (SR1 through SR6 when detectable)
- **Columns:** Time samples

- **Color scale:** -3 to $+3$ z-units
- **Adaptive display:** Dynamically adjusted to the number of harmonics detected per event

2.12.3 Envelopes and Phase Locking Value (PLV)

Per-harmonic envelope and PLV time series were computed as described in Sections 2.5.1 and 2.5.3, providing complementary views of amplitude dynamics and inter-channel phase consistency.

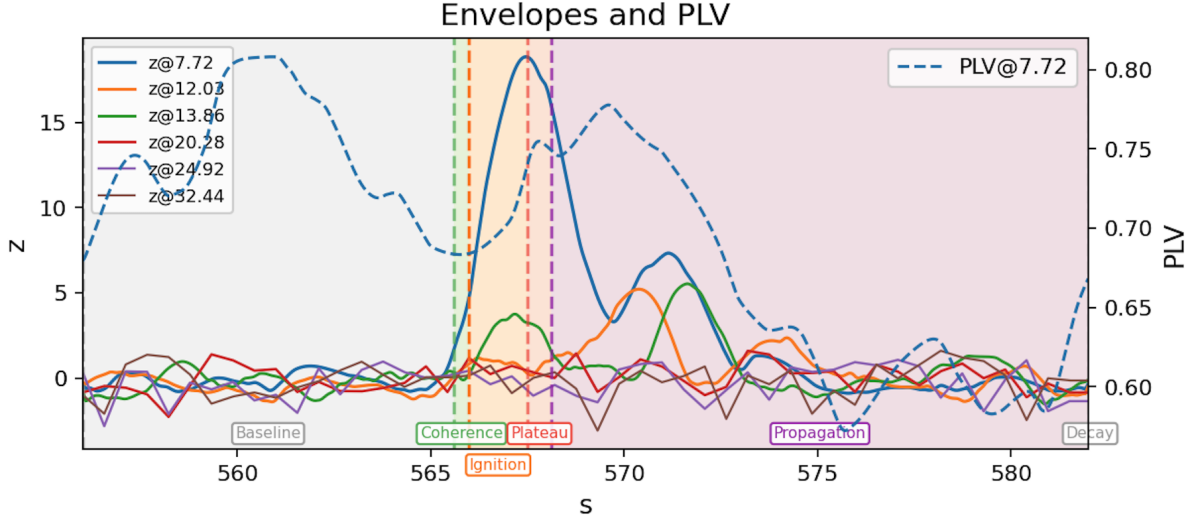


Figure 3: Enter Caption

2.12.4 Harmonic Stack Index (HSI)

The HSI quantifies the ratio of overtone power to fundamental power, providing a measure of harmonic dominance within each event window. The HSI algorithm comprised the following steps:

1. **Fundamental Power:** Bandpass filter at fundamental frequency ($f_0 \pm 0.5$ Hz) and compute mean squared amplitude: $P_f = \langle x_{f_0}^2 \rangle$
2. **Overtone Powers:** For each overtone frequency f_n (SR2 through SR6, excluding the fundamental), bandpass filter and compute mean power: $P_n = \langle x_{f_n}^2 \rangle$
3. **Ratio Computation:** $HSI = \frac{\sum_n P_n}{P_f}$

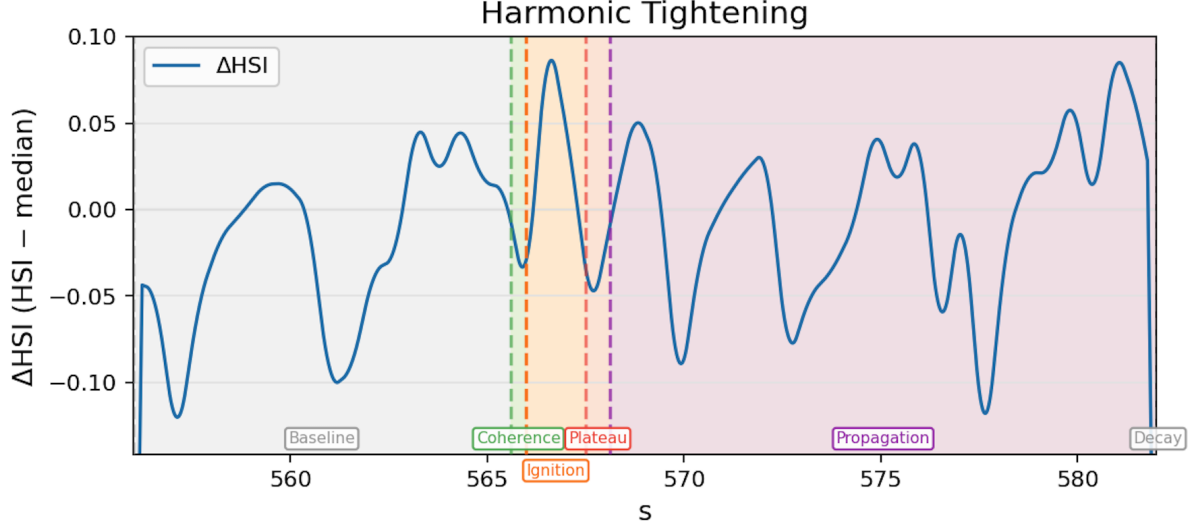


Figure 4: Enter Caption

Higher HSI values indicate overtone-dominated events (harmonics stronger than fundamental); lower HSI values indicate fundamental-dominated events. $\Delta\text{HSI} = \text{HSI}(t) - \text{median}(\text{HSI})$ provided baseline-corrected temporal dynamics, smoothed with 450 ms window. In fundamental-led ignition events, we expect HSI to decrease as the fundamental amplitude increases relative to overtones.

2.12.5 Triadic Bicoherence (Nonlinear Phase Coupling)

Nonlinear phase coupling among SR harmonic triplets was quantified using bicoherence (Kim and Powers, 1979). For frequency triplets (f_1, f_2, f_3) where $f_1 + f_2 \approx f_3$:

$$B(f_1, f_2, f_3; t) = |\langle \exp[i(\phi_1 + \phi_2 - \phi_3)] \rangle|$$

where ϕ_1, ϕ_2, ϕ_3 are instantaneous phases at the respective frequencies, and $\langle \cdot \rangle$ denotes averaging over the analysis window.

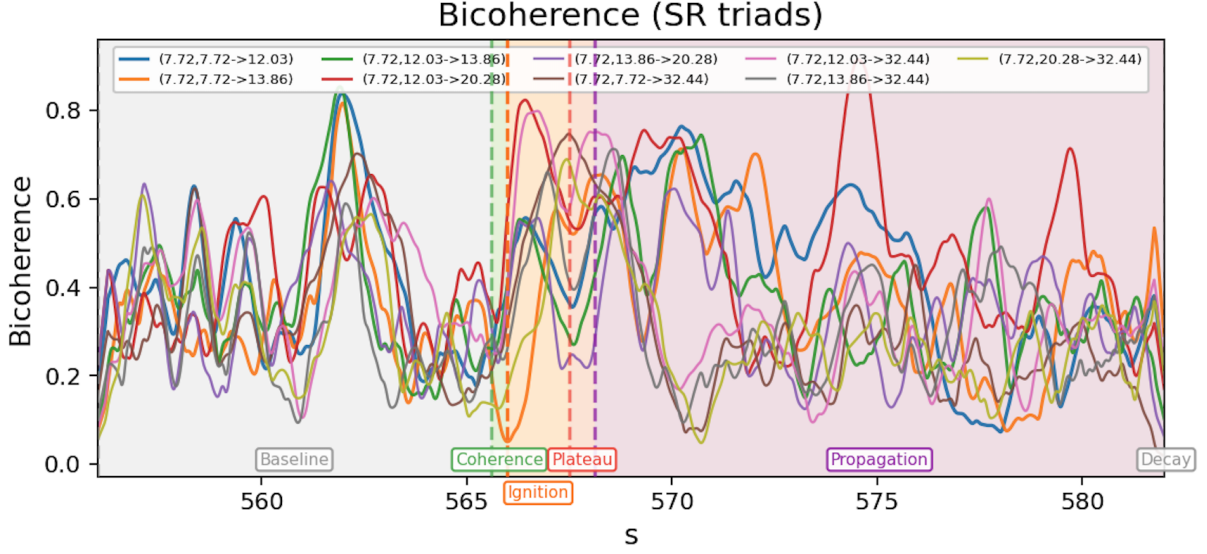


Figure 5: Enter Caption

Analyzed Triads:

- *Fundamental doubling*: (f_1, f_1, f_2) — tests whether $SR1 + SR1 \rightarrow SR2$
- *Consecutive harmonics*: (f_1, f_2, f_3) — tests sequential coupling
- *Non-adjacent couplings*: (f_1, f_3, f_4) — tests higher-order relationships

Parameter	Value	Units	Description
Window duration	0.8	s	Analysis window
Temporal smoothing	500	ms	Smoothing kernel

2.12.6 Theta-Gamma Phase-Amplitude Coupling (PAC)

Cross-frequency coupling between theta phase and gamma amplitude was quantified using the Mean Vector Length (MVL) method (Canolty et al., 2006a; Canolty et al., 2006b; Tort et al., 2010):

- **Phase Signal**: SR fundamental frequency ± 1 Hz, averaged across channels
- **Amplitude Signal**: Gamma band (25–45 Hz), median across channels

Procedure:

1. Extract instantaneous theta phase via Hilbert transform of bandpass-filtered signal
2. Extract gamma amplitude envelope via Hilbert transform
3. Compute MVL in 6-second sliding windows: $MVL = \frac{1}{N} \left| \sum_{t=1}^N A_\gamma(t) \cdot e^{i\phi_\theta(t)} \right|$

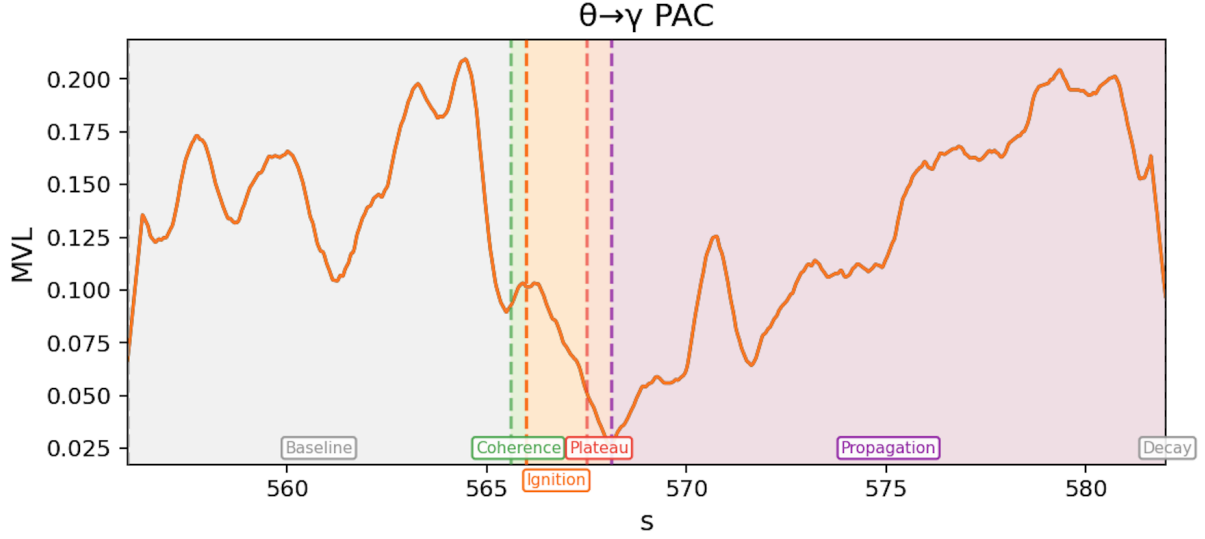


Figure 6: Enter Caption

Gating: Analysis restricted to time points in the top 70th percentile of gamma activity to focus on periods of robust high-frequency oscillations.

2.12.7 Six-Phase Evolution Detection Algorithm

Six temporal phases were identified using a data-driven segmentation algorithm based on threshold crossings and derivative sign changes:

Phase	Name	Detection Criteria
1	Baseline	$PLV \leq 1.1 \times \text{session median}$ AND $z\text{-score} < 2.0$
2	Coherence-First	PLV rising while power remains near baseline (t_0 to $\sim 3s$ post-onset)
3	Power Amplification	Rapid power increase (z rising from 2 to peak)
4	Peak Synchronization	Maximum MSC, PLV, power; $\geq 85\%$ of peak for $\geq 0.5s$
5	Propagation	PAC surge period ($\sim 5s$ window around PAC peak)
6	Decay	Exponential relaxation to baseline

Phase Boundary Criteria:

- Phase 1→2: PLV exceeds $1.1 \times$ baseline median
- Phase 2→3: Power z -score exceeds 2.0
- Phase 3→4: Power derivative becomes negative while power remains $> 85\%$ of maximum
- Phase 4→5: Power drops below 85% of maximum
- Phase 5→6: PLV drops below $1.5 \times$ baseline median
- Phase 6→1: Both power z -score < 2.0 and $PLV \leq 1.1 \times \text{baselinemedian}$

2.13 Network Dynamics and Connectivity Analysis

An alternative analytical framework emphasized network connectivity, information flow topology, and global integration dynamics. These methods complement the event-level characterization (Section 2.12) by revealing inter-electrode coupling patterns and cross-scale information transfer during SIEs.

2.13.1 Harmonic Envelopes

Per-harmonic amplitude envelopes were computed for each electrode using Hilbert transform of bandpass-filtered signals (± 0.5 Hz around each harmonic), providing the input for subsequent connectivity analyses.

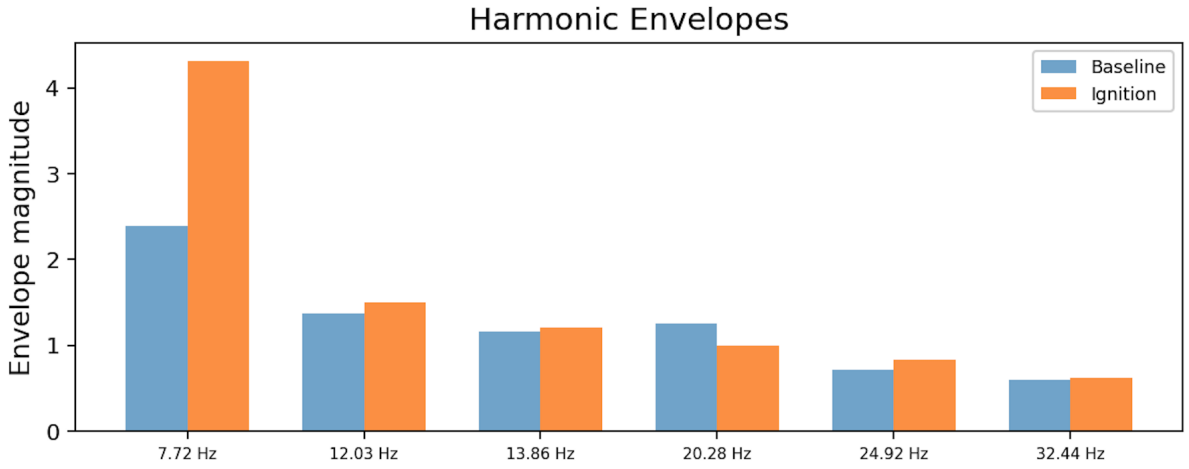


Figure 7: Enter Caption

2.13.2 Frequency-Specific EEG-SR Coherence

Magnitude-squared coherence between each channel and the SR reference signal (median of all channels) was computed for each harmonic:

$$MSC(f) = \frac{|\langle Z_{ch} \cdot \bar{Z}_{ref} \rangle|^2}{\langle |Z_{ch}|^2 \rangle \cdot \langle |Z_{ref}|^2 \rangle}$$

where Z represents the analytic signal at frequency f .

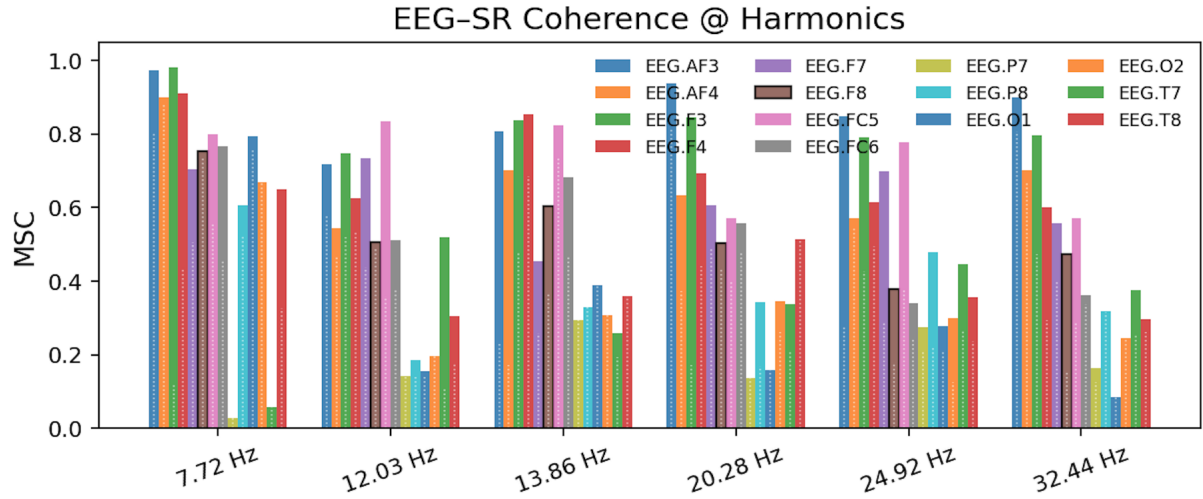


Figure 8: Enter Caption

Statistical Significance:

- *Surrogate testing*: 32 time-shifted reference signals
- *Threshold*: 95th percentile of surrogate distribution

2.13.3 Kuramoto Global Phase Synchronization Dynamics

Full timeseries of the Kuramoto order parameter $R(t)$ was computed at the fundamental frequency:

$$R(t) = \frac{1}{N} \left| \sum_{j=1}^N e^{i\phi_j(t)} \right|$$

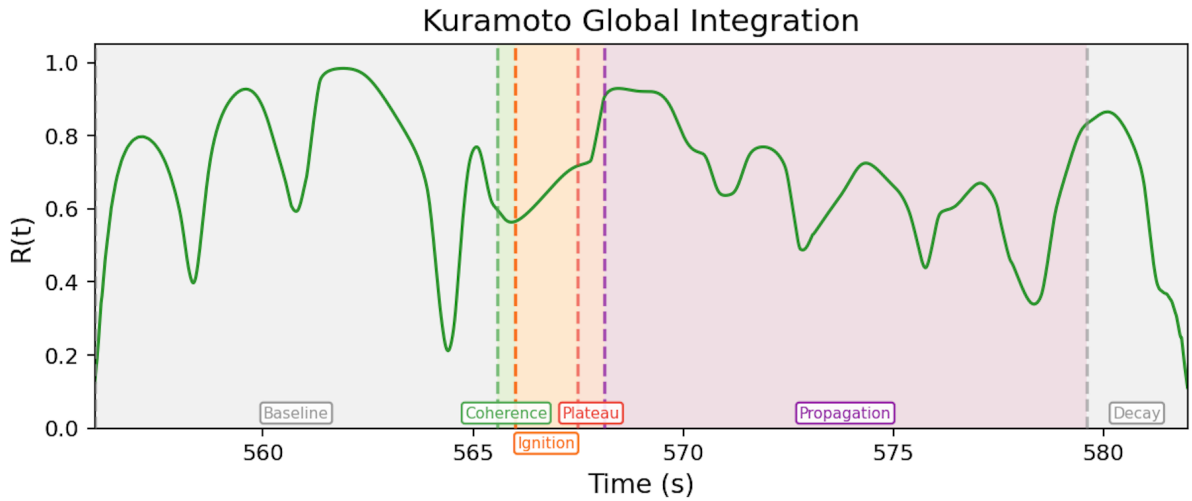


Figure 9: Enter Caption

Parameter	Value	Units	Description
Temporal smoothing	300	ms	Gaussian kernel
Phase overlay	6 phases	—	Color-coded temporal spans

2.13.4 Power-Coherence Phase Space Analysis

Two-dimensional phase portraits were constructed:

- *X-axis*: SR envelope power (log scale)
- *Y-axis*: Kuramoto R
- *Color coding*: Blue = baseline epochs, Orange = ignition epochs

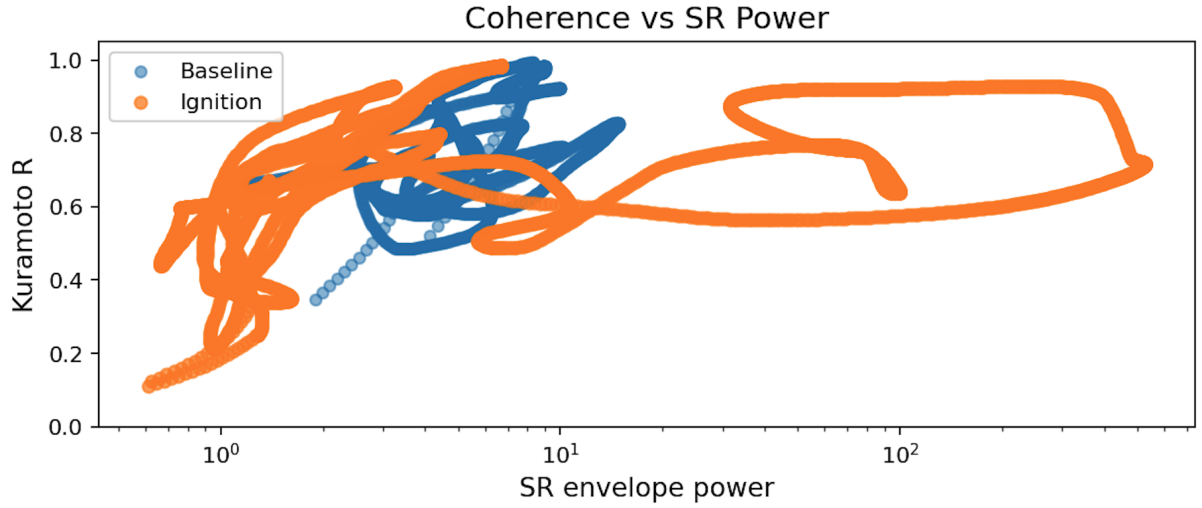


Figure 10: Enter Caption

This visualization reveals:

- Power-law relationships between power and synchronization
- Threshold dynamics (critical R values for state transitions)
- Hysteresis effects (different trajectories for ignition vs. decay)

2.13.5 Directed Information Flow Topology (Transfer Entropy Proxy)

Directed information flow between electrode pairs was quantified using a transfer entropy (Schreiber, 2000) proxy based on lagged correlations:

$$TE(i \rightarrow j) = \rho^2(X_i[t], X_j[t + \tau]) - \rho^2(X_j[t], X_j[t + \tau])$$

where $\tau = 50$ ms (lag), and ρ^2 is squared Pearson correlation.

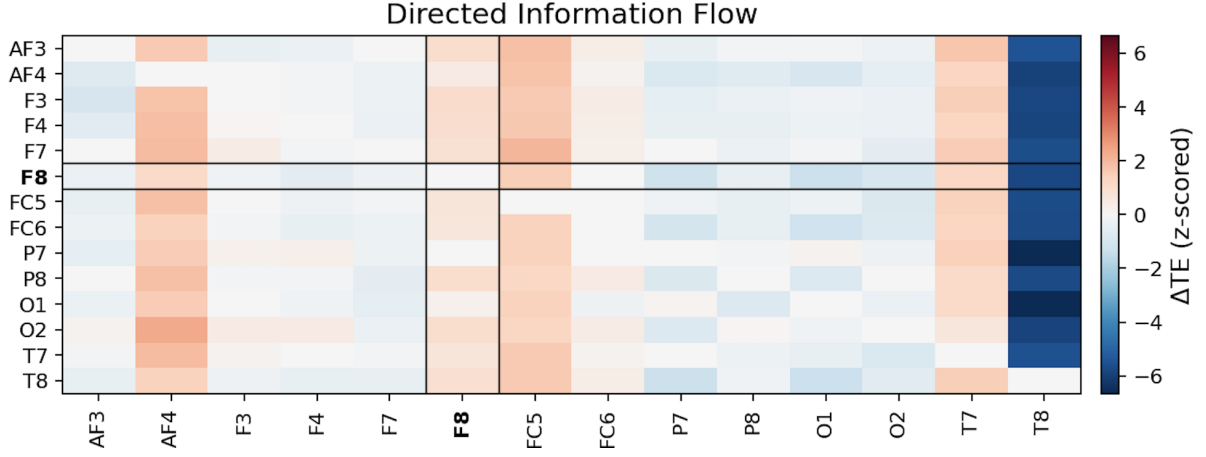


Figure 11: Enter Caption

Baseline Correction: $\Delta TE = TE_{\text{ignition}} - TE_{\text{baseline}}$. Both terms z-scored within session. Positive ΔTE indicates increased directed coupling during SIEs.

Parameter	Value	Units	Description
Lag (τ)	50	ms	Temporal offset
Correlation	ρ^2	—	Squared Pearson

2.13.6 Cross-Scale Information Transfer (Theta→Gamma)

Transfer entropy between theta and gamma envelopes was computed to quantify cross-frequency information flow:

- **Theta envelope:** Fundamental frequency ± 1 Hz, median across channels
- **Gamma envelope:** 30–60 Hz, median across channels

$$TE_{\theta \rightarrow \gamma} = \rho^2(\theta_{\text{env}}[t], \gamma_{\text{env}}[t + 100\text{ms}]) - \rho^2(\gamma_{\text{env}}[t], \gamma_{\text{env}}[t + 100\text{ms}])$$

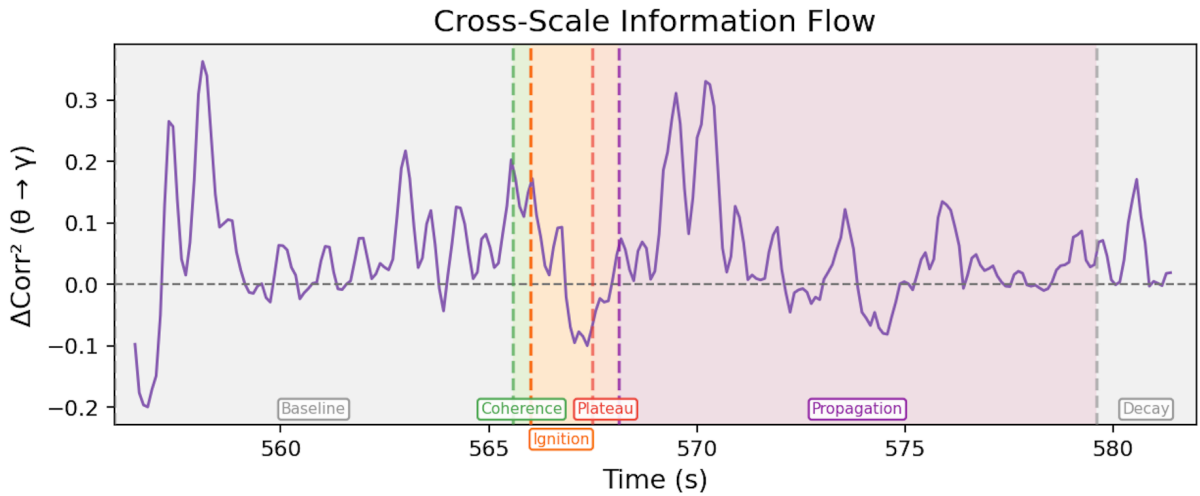


Figure 12: Enter Caption

Parameter	Value	Units	Description
Lag	100	ms	Theta-to-gamma delay
Window duration	1.0	s	Sliding windows
Temporal smoothing	300	ms	Smoothing kernel

2.14 Null Control Analyses

Six null control analyses were implemented to validate that SIE characteristics represent genuine neural organization rather than methodological artifacts, random spectral coincidences, or statistical flukes.

2.14.1 Null Control A: Phase Randomization

Test whether high synchronization values (MSC, PLV, SR-Score) reflect genuine inter-regional phase coupling or arise from spectral content alone.

Procedure:

1. For each detected SIE, extract the multi-channel EEG segment
2. Compute the Fast Fourier Transform (FFT) of each channel
3. Randomize phases by replacing each phase value with a random value drawn uniformly from $[0, 2\pi]$, while preserving the amplitude spectrum exactly
4. Apply inverse FFT to generate phase-shuffled surrogate
5. Repeat steps 2–4 to generate 100 surrogates per event
6. Recompute SR-Score, MSC, and PLV for each surrogate
7. Compare observed values to surrogate distributions

Statistical Assessment:

- Z-score of observed value relative to surrogate distribution
- Percentile rank of observed value
- Cohen’s d comparing observed versus surrogate means

2.14.2 Null Control B: Random Temporal Windows (Baseline Comparison)

Test whether SR-frequency organization is specific to detected SIE events or exists continuously throughout recordings.

Procedure:

1. Identify baseline periods: segments >10 seconds from any detected SIE
2. Randomly sample 3-second windows from baseline periods ($n = 1,366$, matching SIE count)

3. Apply identical FOOOF-based peak extraction within SR harmonic search windows
4. Compare baseline windows to SIE windows on: - Peak detection rate (proportion with detectable peaks in each band) - Peak frequencies (if detected) - Frequency ratios - Power z-scores - MSC and PLV values

Interpretation: If SR frequencies exist only during SIEs, baseline windows should show no SR-band peaks. If SR frequencies are continuous, baseline windows should show peaks but with different quality metrics.

2.14.3 Null Control C: Random Spectral Peak Triplets (Uniform)

Test whether golden ratio precision of SR harmonic frequency ratios is unique or could arise from randomly selected FOOOF peak triplets. This uses uniform sampling from physiologically plausible frequency ranges.

Procedure:

1. Generate 10,000 random frequency triplets by sampling three frequencies uniformly from the 5–40 Hz range
2. For each triplet (f_1, f_2, f_3), compute the three frequency ratios: $f_2/f_1, f_3/f_1, f_3/f_2$
3. Compute the deviation from golden ratio predictions: - Predicted ratios: ϕ (1.618), ϕ^2 (2.618), ϕ^3 (4.236) - Deviation = mean absolute difference between observed and closest ϕ^n value
4. Compare the distribution of random triplet deviations to observed SIE deviation

Statistical Assessment:

- Percentile rank of observed deviation within random distribution
- Cohen’s d between observed deviation and random mean
- Probability of obtaining observed or better precision by chance

Limitation: Uniform sampling from ranges centered on ϕ^n frequencies (e.g., 5–10 Hz centered on $\phi^0 \approx 7.6$ Hz) may bias the null toward producing decent ϕ -ratios. Null Control D addresses this limitation.

2.14.4 Null Control D: Peak-Based Random Triplets

Test whether SIE frequency triplets show better ϕ -convergence than random triplets sampled from the actual distribution of FOOOF peaks in EEG data. This controls for the spectral structure of real EEG recordings.

Procedure:

1. Pool all FOOOF-detected peaks from the complete dataset ($N = 244,955$ peaks from 661 sessions, spanning all cognitive contexts and devices)

2. Filter peaks into SR harmonic bands: SR1 (6.5–9.5 Hz, $n = 23,075$), SR3 (18–22 Hz, $n = 27,659$), SR5 (30–36 Hz, $n = 18,347$)
3. Generate 10,000 random triplets by independently sampling one peak frequency from each band pool
4. Compute composite φ -error for each triplet using the same metric as SIE events
5. Compare SIE φ -error distribution to random peak triplet distribution

Statistical Assessment:

- Permutation test (10,000 iterations)
- Mann-Whitney U test (non-parametric)
- Cohen’s d effect size
- Bootstrap 95% CI for SIE mean φ -error

Interpretation: Unlike the uniform null (Null Control C), the peak-based null directly tests: “Given that EEG spectra naturally contain peaks at various frequencies, do SIE events select frequency combinations that are more φ -aligned than typical?” A significant result indicates that SIE detection preferentially identifies genuinely exceptional φ^n organization.

2.14.5 Null Control E: Shuffled Bootstrap (Within-Event Coordination)

Test whether harmonic frequency relationships arise from event-level coordination (frequencies within an event are coupled) or from population-level marginal distributions (each frequency follows an independent distribution that happens to produce good ratios).

Procedure:

1. Extract the three harmonic frequencies (SR1, SR3, SR5) from all 1,366 events
2. Create shuffled datasets by independently permuting each harmonic frequency vector: - Shuffle SR1 frequencies across events (random assignment) - Shuffle SR3 frequencies across events (independent of SR1 shuffle) - Shuffle SR5 frequencies across events (independent of SR1, SR3 shuffles)
3. This preserves the marginal distribution of each frequency while destroying within-event pairings
4. Compute the composite φ -error (mean absolute deviation from golden ratio predictions across all three ratios) for each shuffled dataset
5. Repeat 1,000 times to generate null distribution
6. Compare observed composite φ -error to null distribution

Interpretation: If observed error is significantly lower than shuffled error, within-event frequency coordination contributes to precision beyond marginal distributions alone.

2.14.6 Null Control F: Distributional Null Model

Quantify the expected precision arising from the marginal frequency distributions alone, providing a more precise baseline for evaluating within-event effects.

Procedure:

1. Model each harmonic frequency as a univariate normal distribution using observed means and standard deviations
2. Generate 10,000 synthetic datasets by independently sampling from each distribution
3. Compute frequency ratios and ϕ -error for each synthetic dataset
4. Compare observed error to the distribution of synthetic errors

Interpretation: Similarity between observed and synthetic precision suggests that marginal distributions largely determine ratio precision; substantial improvement suggests additional within-event coordination.

2.15 Cross-Validation Analyses

2.15.1 Device Independence

To test whether SIE characteristics are device-independent, one-way ANOVA was performed comparing events from Muse ($n = 165$), Emotiv EPOC X ($n = 1,144$), and Emotiv Insight ($n = 57$) devices on all primary metrics (harmonic frequencies, MSC, PLV, SR-Score).

2.15.2 Context Independence

To test whether SIEs differ between meditation and cognitive task contexts, independent-samples t-tests compared meditation ($n = 260$) versus PhySF ($n = 447$) events on all primary metrics.

2.15.3 Task Condition Comparison

Within the PhySF dataset, paired-samples t-tests compared flow versus non-flow conditions on SIE detection rate (events per session) and event characteristics.

2.16 Golden Ratio Analysis

An unexpected observation during initial characterization was that SIE harmonic frequencies approximate powers of the golden ratio ($\phi = 1.6180339\dots$) (Livio, 2002). To quantify this relationship:

2.16.1 Predicted Frequencies

Analysis of detected SIE events reveals that harmonic frequencies follow powers of the golden ratio scaled by the fundamental frequency:

$$f_n = f_0 \cdot \phi^n \tag{1}$$

where $f_0 = 7.6$ Hz is the Schumann fundamental frequency determined from SR observatory data, and n is an integer representing harmonic level.

This empirical relationship can be connected to physical theory by assuming $f_0 = \frac{c}{2\pi r}$, where c is the speed of light (299,792,458 m/s) and r is Earth's radius (6,367,000 m). Converting to angular frequency $\omega = 2\pi f$:

$$\omega_n = 2\pi f_0 \cdot \varphi^n = 2\pi \cdot \frac{c}{2\pi r} \cdot \varphi^n = \frac{c}{r} \cdot \varphi^n \quad (2)$$

Using $f_0 = 7.6$ Hz, the predicted neural frequencies are:

n	φ^n	Predicted Frequency (Hz)
0	1.000	7.60 (fundamental)
2	2.618	19.90 (SR3)
3	4.236	32.19 (SR5)

2.16.2 Ratio Precision Analysis

For each SIE event, three frequency ratios were computed:

- SR3/SR1 (predicted: $\varphi^2 = 2.618$)
- SR5/SR1 (predicted: $\varphi^3 = 4.236$)
- SR5/SR3 (predicted: $\varphi = 1.618$)

Percentage error was computed as: $|\text{observed} - \text{predicted}| / \text{predicted} \times 100$

2.16.3 Independence-Convergence Analysis

To test whether individual frequencies vary independently while ratios converge, we computed:

1. Pairwise Pearson correlations between SR1, SR3, and SR5 frequencies across events
2. The coefficient of variation (CV) for each frequency and each ratio
3. Scatter plots of frequency pairs with theoretical prediction lines

2.17 Statistical Analysis

2.17.1 General Approach

All statistical tests were two-tailed with $\alpha = 0.05$. Effect sizes are reported as Cohen's d for t-tests (small: 0.2, medium: 0.5, large: 0.8) and Pearson's r for correlations.

2.17.2 Multiple Comparisons

For analyses involving multiple comparisons (e.g., three harmonic frequencies $\times$ multiple metrics), Bonferroni correction was applied where indicated.

2.17.3 Non-Parametric Alternatives

When distributional assumptions were violated (Shapiro-Wilk $p < 0.05$), non-parametric alternatives were employed: Mann-Whitney U for independent samples, Wilcoxon signed-rank for paired samples, Kruskal-Wallis for multiple groups.

2.17.4 Bootstrap Confidence Intervals

For derived metrics without known sampling distributions (e.g., composite ϕ -error), 95% confidence intervals were estimated via percentile bootstrap with 10,000 resamples.

2.18 Data and Code Availability

2.18.1 Meditation Dataset

Longitudinal meditation recordings will be made available upon publication via:

- <https://github.com/neurokinetikz/schumann>.

2.18.2 Analysis Code

Complete Python analysis code, including the SIE detection pipeline, null control implementations, and statistical analyses, will be made available upon publication at:

- <https://github.com/neurokinetikz/schumann>.

2.18.3 Derived Data

Event-level data for all 1,366 detected SIEs, including harmonic frequencies, synchronization metrics, and quality scores, will be provided as supplementary material.

2.19 Declarations

2.19.1 Conflicts of Interest

The authors declare no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

2.19.2 Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

2.20 Software and Computing Environment

All analyses were performed on a MacBook Air (Apple M4, 16 GB RAM) running macOS Tahoe 26.1 using the following software environment:

Software	Version	Purpose
Python	3.9.7	Primary analysis environment
NumPy (Harris et al., 2020)	1.21.0	Numerical computing
SciPy (Virtanen et al., 2020)	1.7.1	Scientific computing, signal processing
MNE-Python (Gramfort et al., 2013)	0.24.0	EEG data handling, spectral analysis
FOOOF (Donoghue et al., 2020)	1.0.0	Spectral parameterization
Matplotlib (Hunter, 2007)	3.4.3	Visualization
Pandas (McKinney, 2010)	1.3.3	Data management

3 Results

3.1 Dataset Overview and Event Detection

Application of the seven-stage SIE detection pipeline to the combined dataset yielded 1,366 discrete events across 661 recording sessions from 91 unique participants (Table 2). Events were detected across all six datasets and five cognitive contexts, with overall mean detection rate of 2.1 ± 3.3 events per session (range: 1–27; individual dataset means ranged from 1.1 to 9.7 events per session). Event duration averaged 26.9 ± 21.3 seconds (median: 20.0 seconds; minimum window size: 20 seconds), consistent with the transient, high-coherence nature of SIE phenomena and characteristic timescales of transient oscillatory events (Siegel et al., 2012).

Dataset	Participants	Sessions	SIE Events	Events/Session	Device(s)	Context
PhySF	25	46	447	9.7 ± 6.2	EPOC X	Flow/Non-flow
Meditation	1	34	260	7.6 ± 4.8	Muse, EPOC X, Insight	Eyes-closed
MultiPENG	36	533	604	1.1 ± 0.4	EPOC X	Gaming
VEP	29	48	55	1.1 ± 0.5	EPOC X	Visual
Combined	91	661	1,366	2.1 ± 3.3	3 types	5 contexts

The distribution of events across datasets revealed that PhySF cognitive flow recordings and meditation recordings yielded the highest per-session detection rates (9.7 and 7.6 events/session respectively), with gaming and visual perception tasks showing lower rates (~ 1 event/session). This pattern may reflect differential engagement of SR-aligned neural mechanisms across cognitive states, though recording duration differences may also contribute.

3.2 Harmonic Definitions

Spectral analysis of detected SIEs revealed nine distinct harmonic frequencies spanning the 7–41 Hz range (Table 4). These harmonics, labeled SR1 through SR6 (with half-integer labels SR1.5, SR2.5, and quarter-integer SR2o), serve as the basis for all subsequent analyses. The fundamental frequency (SR1) centers near 7.6 Hz at the theta/alpha boundary, with higher harmonics extending through the alpha, beta, and gamma bands.

Table 4. Harmonic Frequency Definitions

Harmonic	Measured (Hz)	CV (%)	N	EEG Band
SR1	7.63 ± 0.33	3.18	1121	θ/α Boundary
SR1.5	9.96 ± 0.35	3.56	1139	α Attractor
SR2	11.98 ± 0.40	3.37	1050	α/β Boundary
SR2o	13.76 ± 0.44	3.17	1150	Low β
SR2.5	15.50 ± 0.46	2.99	1188	β Attractor
SR3	19.99 ± 0.57	2.02	1250	β Boundary
SR4	24.98 ± 1.17	4.69	1359	High β Attractor
SR5	32.57 ± 1.23	1.79	1364	β/γ Boundary
SR6	40.11 ± 1.61	4.01	1365	γ Attractor

CV = coefficient of variation. N = events with detectable harmonic. EEG band assignments follow canonical frequency definitions (Buzsáki and Draguhn, 2004a).

3.3 Exemplar Event Walkthrough

To illustrate these harmonics in action, we present a detailed analysis of a canonical high-quality SIE event before examining population-level statistics.

Figure 3 presents a canonical high-quality SIE event (SR-score = 1.04, top 1.5% of events) recorded during eyes-closed meditation using the Emotiv EPOC X device. This 26-second event (556–582s) illustrates the complete temporal dynamics and multi-scale organization characteristic of SIE phenomena. Each panel reveals distinct aspects of the ignition process. It should be noted that while this exemplar exhibits the general characteristics of SIE phenomena, individual events vary considerably in their specific dynamics, timing, and magnitude of effects; the population-level statistics presented subsequently capture this variability.

3.3.1 Spectral and Temporal Dynamics

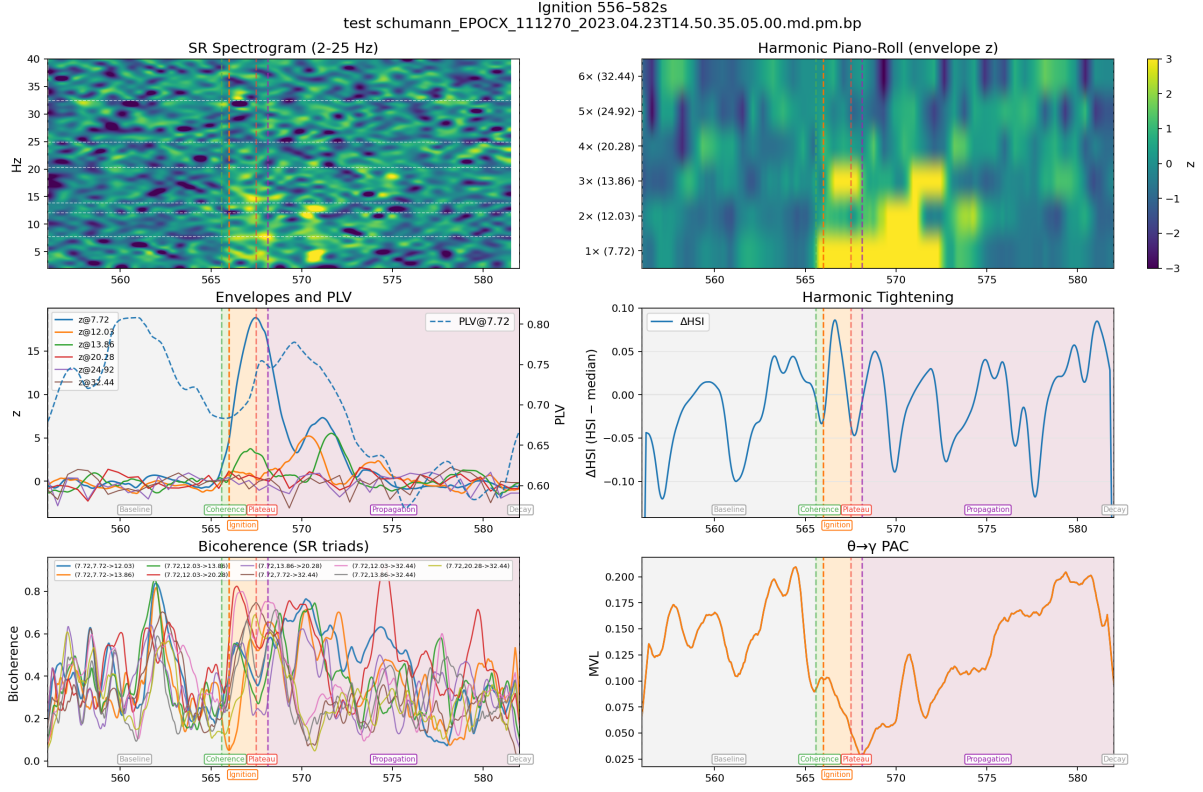


Figure 13: Exemplar SIE event spectral and temporal dynamics. (A) Time-frequency spectrogram showing harmonic activation at φ^n frequencies. (B) Harmonic piano-roll showing z-score elevation across all six harmonics. (C) Envelope traces overlaid with phase-locking value, demonstrating coherence-first dynamics. (D) Harmonic Stack Index showing fundamental dominance (decreasing HSI) preceding ignition.

Panel A: SR Spectrogram (2–40 Hz). The time-frequency representation reveals broad-band spectral changes during ignition. Horizontal dashed lines mark the predicted φ^n harmonic frequencies (7.72, 12.03, 13.86, 20.28, 24.92, 32.44 Hz). Key observations:

- **Pre-ignition (556–564s):** Diffuse spectral activity without clear harmonic organization. Power distributed across frequencies with no preferential concentration at φ^n positions.
- **Coherence/Plateau phase (564–568s):** Emergence of distinct horizontal "bands" of elevated power precisely aligned to the dashed harmonic markers. This alignment confirms that power increases occur specifically at φ^n frequencies rather than broadband.
- **Ignition (568–570s):** Maximal power concentration at harmonic frequencies, visible as bright yellow horizontal striations. The fundamental (7.72 Hz) and second harmonic (12.03 Hz) show strongest activation.
- **Propagation (570–578s):** Sustained but gradually diminishing harmonic structure. Higher harmonics (20–32 Hz) maintain elevation longer than would be expected from simple theta-band filtering artifacts.

Panel B: Harmonic Piano-Roll (envelope z-score). The piano-roll representation isolates activity at each φ^n harmonic, with z-score color mapping (-3 to $+3\sigma$) referenced to session baseline. Y-axis labels show harmonic order and measured frequency: $1\times$ (7.72 Hz), $2\times$ (12.03 Hz), $3\times$ (13.86 Hz), $4\times$ (20.28 Hz), $5\times$ (24.92 Hz), $6\times$ (32.44 Hz).

Critical observations:

- **Multi-band activation:** All six harmonics show simultaneous $z > 2\sigma$ elevation during ignition, confirming this is not merely theta-band activity bleeding into higher frequencies.
- **Activation sequence:** The fundamental ($1\times$) activates first (~ 565 s), followed by sequential activation of higher harmonics over ~ 2 seconds. This bottom-up cascade suggests hierarchical harmonic generation rather than independent oscillators.
- **Differential persistence:** Lower harmonics ($1\times$, $2\times$) show sustained elevation throughout propagation (570–578s), while higher harmonics ($5\times$, $6\times$) return to baseline more rapidly. This temporal dissociation argues against simple spectral leakage.
- **Peak z-scores:** Fundamental shows $z > 3$ (yellow), while $6\times$ harmonic shows $z \approx 1.5$ – 2 (green-yellow), consistent with the population gradient (SR1: 97.8% $> 3\sigma$; SR6: 40.7% $> 3\sigma$).

Panel C: Envelopes and PLV. This panel overlays z-score envelopes at each harmonic (solid colored lines) with phase-locking value at the fundamental frequency (blue dashed line, right y-axis). The six-phase temporal structure is annotated at bottom.

The coherence-first signature:

- **PLV rises first:** Phase-locking begins increasing at ~ 562 s, reaching peak (~ 0.82) at ~ 566 s—approximately 2–3 seconds BEFORE z-score peaks.
- **Z-scores follow:** Power envelopes remain near baseline until ~ 565 s, then rise sharply to peak at ~ 568 s.
- **Temporal offset:** This ~ 2 – 3 second lag between coherence and amplitude is a hallmark of network-level coordination, inconsistent with local oscillatory bursts where amplitude and phase would co-vary.

Panel D: Fundamental Dominance (Δ HSI). The Harmonic Stack Index deviation from session median tracks the ratio of overtone power to fundamental power. Negative Δ HSI indicates the fundamental is stronger relative to overtones than typical (fundamental-dominated); positive Δ HSI indicates overtone-dominated activity.

Key pattern:

- **Baseline:** Δ HSI fluctuates around 0 to $+0.05$ (typical overtone/fundamental balance).
- **Pre-ignition shift:** Δ HSI begins decreasing at ~ 563 s, reaching minimum (-0.10) during the plateau phase—BEFORE peak power elevation. This indicates the fundamental (SR1) is gaining power relative to overtones.

- **Fundamental dominance precedes ignition:** The shift toward fundamental-led activity is established ~ 3 seconds before maximal amplitude, suggesting that SR1 dominance is a prerequisite for, not consequence of, ignition.
- **Relaxation during propagation:** ΔHSI returns toward zero during propagation, indicating gradual return to typical overtone/fundamental balance.

This temporal sequence (fundamental dominance $\rightarrow$ coherence $\rightarrow$ amplitude) strongly supports a model where SR1-led frequency organization must be established before power can increase.

3.3.2 Cross-Frequency Coupling

Panel E: Bicoherence (SR Triads). Bicoherence quantifies nonlinear (quadratic) phase coupling between frequency triplets. Each trace represents a specific triad: $(f_1, f_2 \rightarrow f_1+f_2)$. Legends show triads including $(7.72, 7.72 \rightarrow 12.03)$, $(7.72, 12.03 \rightarrow 13.86)$, $(7.72, 12.03 \rightarrow 20.28)$, etc.

Observations:

- **Baseline bicoherence:** ~ 0.2 – 0.4 across triads, indicating modest nonlinear coupling during normal activity.
- **Ignition-phase surge:** Multiple triads simultaneously increase to bicoherence > 0.8 during the coherence/plateau phases (~ 565 – 570 s). This coordination across triads indicates that nonlinear harmonic generation is not frequency-specific but affects the entire φ^n stack.
- **Triad synchrony:** The remarkable synchronization of bicoherence increases across different frequency combinations suggests a common generative mechanism rather than independent oscillator coupling.
- **Peak timing:** Bicoherence peaks slightly BEFORE maximal z-scores, consistent with nonlinear coupling driving harmonic power rather than vice versa.

Panel F: $\vartheta \rightarrow \gamma$ Phase-Amplitude Coupling (PAC). Mean Vector Length (MVL) quantifies the modulation of gamma-band (30–45 Hz) amplitude by theta-band (4–8 Hz) phase—a hallmark of hierarchical neural coding (J. E. Lisman and Jensen, 2013). Higher MVL indicates stronger cross-frequency coupling.

Temporal profile:

- **Baseline (556–563s):** MVL ~ 0.05 – 0.08 , indicating weak theta-gamma coupling.
- **Pre-ignition rise (563–566s):** MVL begins increasing, reaching ~ 0.15 before z-score elevation.
- **Peak PAC (566–568s):** MVL reaches maximum (~ 0.20) during the plateau phase, representing a $\sim 3\times$ increase from baseline.
- **Sustained coupling (568–577s):** MVL remains elevated (~ 0.12 – 0.15) throughout propagation.

- **Decay (577–582s):** Gradual return to baseline levels.

The theta-gamma coupling increase precedes and accompanies ignition, suggesting that cross-frequency phase relationships are integral to SIE dynamics. The $\sim 3\times$ increase in MVL during ignition indicates substantial enhancement of hierarchical frequency organization.

3.3.3 Network Synchronization

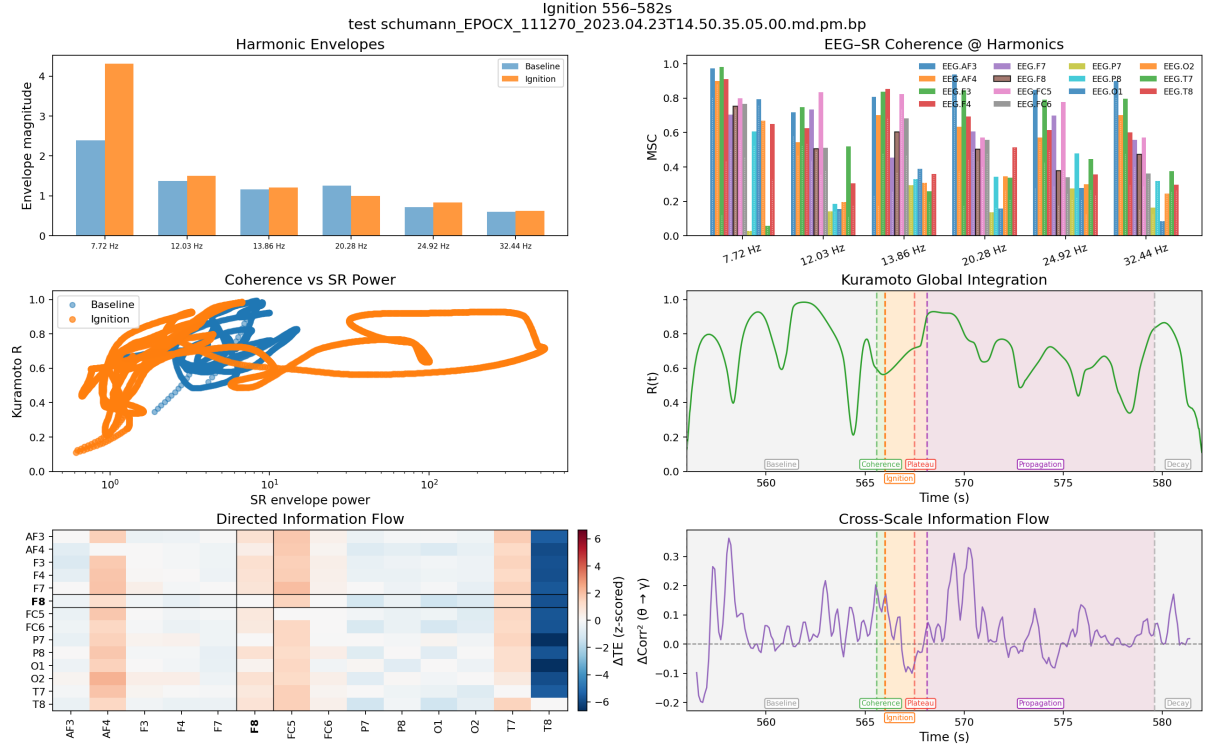


Figure 14: Exemplar SIE event network dynamics. (G) Harmonic envelope comparison between baseline and ignition states. (H) EEG-SR coherence by electrode and harmonic. (I) State-space trajectory showing coherence vs. power. (J) Kuramoto order parameter time series. (K) Transfer entropy matrix. (L) Cross-scale theta-gamma correlation dynamics.

Panel G: Harmonic Envelopes (Baseline vs. Ignition). Bar chart comparing mean envelope magnitude during baseline (blue) versus ignition (orange) periods at each harmonic frequency.

Harmonic	Baseline	Ignition	Ratio
7.72 Hz (SR1)	~ 2.4	~ 4.5	1.88$\times$
12.03 Hz (SR2)	~ 1.4	~ 1.5	1.07 $\times$
13.86 Hz (SR2o)	~ 1.2	~ 1.5	1.25 $\times$
20.28 Hz (SR3)	~ 1.2	~ 1.3	1.08 $\times$
24.92 Hz (SR4)	~ 0.7	~ 0.8	1.14 $\times$
32.44 Hz (SR5)	~ 0.6	~ 0.7	1.17 $\times$

The fundamental frequency shows dramatically greater ignition/baseline ratio ($1.88\times$) compared to higher harmonics ($1.07\text{--}1.25\times$), confirming that SR1 is the primary driver of ignition events.

Panel H: EEG-SR Coherence at Harmonics (by Electrode). Bar chart showing magnitude-squared coherence (MSC) between each of 14 EEG channels and a synthetic SR reference signal, computed separately for each harmonic frequency. All 14 electrodes show $\text{MSC} > 0.8$ at SR1 (7.72 Hz), indicating whole-brain synchronization at the fundamental frequency. MSC systematically decreases at higher harmonics, from $\sim 0.85\text{--}0.95$ at SR1 to $\sim 0.30\text{--}0.50$ at SR5. All electrode regions (frontal, temporal, parietal, occipital) show similar patterns, arguing against localized generators.

Panel I: Coherence vs. SR Power State Space. Scatter plot showing the trajectory through coherence (Kuramoto R , y-axis) versus SR envelope power (x-axis, log scale) during baseline (blue) and ignition (orange). Baseline points cluster at low power ($\sim 1\text{--}5$) and moderate coherence ($R \sim 0.5\text{--}0.7$). The ignition trajectory extends rightward to power > 100 while maintaining or increasing coherence ($R \sim 0.7\text{--}0.95$), forming a characteristic "lasso" pattern. The clear non-overlap between baseline and ignition states demonstrates that SIEs represent a distinct neural regime.

Panel J: Kuramoto Global Integration $R(t)$. Time series of the Kuramoto order parameter $R(t)$, measuring instantaneous global phase synchronization across all electrodes. The six-phase temporal structure is visible:

1. **Baseline (556–563s):** R fluctuates between $0.4\text{--}0.7$.
2. **Coherence rise (563–566s):** R increases sharply from ~ 0.5 to ~ 0.85 .
3. **Plateau (566–568s):** R stabilizes at $\sim 0.90\text{--}0.95$ —near-maximal global synchronization.
4. **Ignition (568s):** Brief fluctuation with peak power.
5. **Propagation (568–577s):** R remains elevated ($0.70\text{--}0.90$).
6. **Decay (577–582s):** Gradual decline to baseline.

The plateau at $R > 0.9$ indicates that $> 90\%$ of electrode pairs are phase-synchronized—a highly organized state rarely sustained for multiple seconds.

3.3.4 Information Flow Topology

Panel K: Directed Information Flow (Transfer Entropy Matrix). Asymmetric matrix showing z-scored change in transfer entropy (ΔTE) between all electrode pairs during ignition relative to baseline. Frontal sources (AF3, AF4, F3, F4) show strong positive ΔTE ($z > 4$) to most targets, indicating these regions become information sources during ignition. F8 shows particularly strong outward connections ($z > 6$), suggesting right frontal dominance in information broadcast. Parietal (P7, P8) and occipital (O1, O2) regions primarily receive information, consistent with frontal-to-posterior information flow.

Panel L: Cross-Scale Information Flow ($\Delta\text{corr}^2 \vartheta \rightarrow \gamma$). Time series showing the change in squared correlation between theta-band (4–8 Hz) and gamma-band (30–45 Hz) activity. A biphasic pattern emerges: pre-ignition surge ($\sim +0.35$ at 563–566s) indicates sudden theta-gamma coupling, followed by ignition-phase reversal (~ -0.15 at 566–570s) suggesting decorrelation during maximal activity. This pattern suggests theta-gamma coupling initially increases to establish cross-frequency coordination, then partially decouples to allow independent processing before re-coupling during propagation.

3.3.5 Exemplar Event Summary

This high-quality SIE event demonstrates the complete set of signatures characteristic of the phenomenon:

Feature	Observation	Implication
Multi-band activation	All 6 φ^n harmonics show $z > 2\sigma$	Not theta-only phenomenon
Coherence-first	PLV peaks 2–3s before z-scores	Network coordination precedes amplitude
Fundamental dominance	ΔHSI minimum precedes ignition	SR1-led organization established first
Bicoherence surge	Multiple triads reach > 0.8	Nonlinear harmonic coupling
$\theta \rightarrow \gamma$ PAC increase	MVL triples during ignition	Cross-frequency phase coupling
Global synchronization	Kuramoto $R > 0.9$ at plateau	Whole-brain phase alignment
Frontal hub topology	F8, AF3/4 show elevated ΔTE	Directed information broadcast
State-space separation	Distinct ignition vs. baseline clusters	Qualitatively different neural state

The temporal sequence—fundamental dominance $\rightarrow$ theta-gamma coupling $\rightarrow$ coherence rise $\rightarrow$ power elevation $\rightarrow$ propagation $\rightarrow$ coordinated decay—suggests a hierarchical organization process that must be established before ignition can occur.

Having established the characteristic signatures of individual events, we now quantify these properties across the full population of 1,366 detected SIEs.

3.4 Statistical Validation

The preceding characterization of SIE events raises the question of whether these patterns reflect genuine neural organization or statistical artifacts. We address this through multiple complementary validation approaches.

3.4.1 Cross-Device Validation

One-way ANOVA tested whether harmonic frequencies differed across the three EEG recording devices (Emotiv EPOC X, $n = 1,144$; Muse, $n = 165$; Emotiv Insight, $n = 57$). Device effects varied by harmonic (Table 9): SR1 and SR3 showed no significant differences ($p > 0.25$), while SR5 exhibited a significant device effect ($F = 5.74$, $p = 0.003$), with Muse devices detecting systematically higher frequencies (32.88 Hz) than Emotiv devices (32.53–32.54 Hz).

Harmonic	EPOC X (Hz)	Muse (Hz)	Insight (Hz)	F	p
SR1	7.63 ± 0.33	7.66 ± 0.32	7.68 ± 0.32	1.26	0.28
SR3	19.98 ± 0.57	20.03 ± 0.57	19.88 ± 0.52	1.34	0.26
SR5	32.53 ± 1.23	32.88 ± 1.08	32.54 ± 1.41	5.74	0.003

For SR1 and SR3, cross-device frequency convergence (means within 0.3% of grand mean) supports biological rather than device-specific origins. The significant SR5 device effect likely reflects differential hardware filtering or spectral sensitivity at higher frequencies (32 Hz), where consumer-grade devices show greater variability. Importantly, the core ϕ^n ratios remain preserved across devices: the fundamental (SR1) and first ratio-determining harmonic (SR3) show device-independent detection. This validation is particularly notable given substantial hardware differences: channel count ranged from 4 (Muse) to 14 (EPOC X), electrode types included dry (Muse), semi-dry polymer (Insight), and saline-soaked felt (EPOC X), and spatial coverage varied from frontal-temporal (Muse) to comprehensive scalp coverage (EPOC X).

3.4.2 Context Independence

One-way ANOVA across five cognitive contexts (meditation, flow, non-flow, gaming, visual) revealed no significant frequency differences for any primary harmonic (Table 10).

Harmonic	Meditation	Flow	Non-Flow	Gaming	Visual	F	p
SR1 (Hz)	7.67	7.66	7.62	7.61	7.61	1.46	0.21
SR3 (Hz)	19.98	19.99	20.01	19.98	19.95	0.22	0.93
SR5 (Hz)	32.74	32.56	32.57	32.52	32.40	1.68	0.15

The ϕ^n architecture is preserved across contexts spanning eyes-closed meditation, cognitively demanding flow states, gaming engagement, and passive visual perception. This cross-context stability supports the interpretation that SIE frequencies reflect fundamental neural organizational constraints rather than task-specific processing.

3.4.3 Null Control Validation

To evaluate whether observed ϕ^n precision is spectrally exceptional, we compared SIE harmonic triplets against random frequency triplets using two complementary null models.

Null Model A: Uniform Sampling. We generated 10,000 random frequency triplets by sampling uniformly from physiologically plausible ranges (SR1: 5–10 Hz; SR3: 17–23 Hz; SR5: 28–38 Hz). This yielded:

- Observed mean ϕ -error: 4.37%
- Random triplet mean error: $11.5\% \pm 4.1\%$
- Cohen’s d: 1.71 (very large effect)
- p-value: 0.074

The marginal p-value (0.074) reflects a limitation of this null model: the sampling ranges are centered on φ^n frequencies by construction ($7.5 \text{ Hz} \approx \varphi^0$, $20 \text{ Hz} \approx \varphi^2$, $33 \text{ Hz} \approx \varphi^3$), biasing the null toward producing decent φ -ratios.

Null Model B: Peak-Based Sampling. To address this limitation, we implemented a more rigorous test using actual FOOF peaks detected across all EEG sessions ($N = 244,955$ peaks from 661 recordings). Peaks were filtered into SR bands (SR1: 6.5–9.5 Hz, $n = 23,075$; SR3: 18–22 Hz, $n = 27,659$; SR5: 30–36 Hz, $n = 18,347$), and 10,000 random triplets were generated by sampling one peak from each pool. This controls for the actual spectral structure of EEG data rather than assuming uniform distributions.

Peak-Based Results:

- Observed mean φ -error: $4.37\% \pm 2.00\%$
- Random peak triplet mean error: $9.05\% \pm 4.15\%$
- Cohen’s d: 1.44 (large effect)
- p-value: < 0.0001 (permutation test)
- p-value: < 0.0001 (Mann-Whitney U)

The peak-based null demonstrates that SIE events show **significantly better φ -precision** than random triplets sampled from actual EEG spectral peaks. The highly significant result ($p < 0.0001$) with a large effect size ($d = 1.44$) indicates that the SIE detection algorithm preferentially identifies frequency combinations exhibiting exceptional φ^n organization.

Null Model	p-value	Cohen’s d	Interpretation
Uniform sampling	0.074	1.71	Marginal (biased null)
Peak-based	< 0.0001	1.44	Highly significant

Table 3: Comparison of null control approaches for φ^n precision validation.

Peak-Based Null Control: SIGNIFICANT ($p < 0.05$)

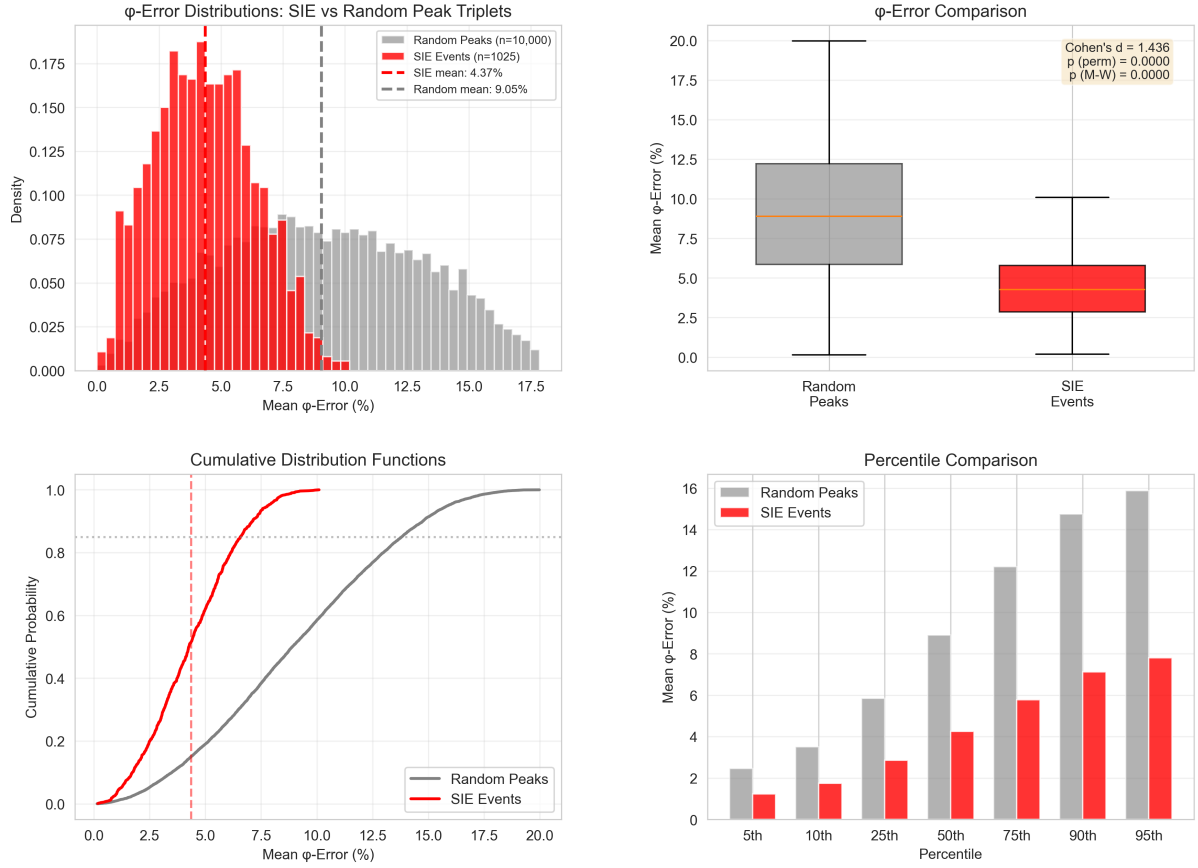


Figure 15: Peak-based null control results. (A) Histogram comparing ϕ -error distributions for SIE events (red) versus random triplets sampled from actual FOOOF peaks (gray). SIE events cluster at lower ϕ -error values. (B) Box plot comparison showing significantly lower ϕ -error for SIE events. (C) Cumulative distribution functions demonstrating separation between distributions. (D) Percentile comparison across the error distribution.

3.4.4 Subject-Level Reproducibility

To verify that findings are not driven by a subset of participants, we computed subject-level φ^n ratio precision ($N = 91$ subjects).

Subject-Level Results:

- Mean SR3/SR1 ratio: 2.6254 ± 0.066 (predicted: 2.6180; error: +0.28%)
- Mean SR5/SR3 ratio: 1.6270 ± 0.045 (predicted: 1.6180; error: +0.56%)

Subject-level ratio errors remained below 1% for both primary ratios, confirming that φ^n precision is not an artifact of averaging across heterogeneous subjects but represents a consistent organizational principle present within individual participants.

3.5 Synchronization Metrics

Having validated the robustness and specificity of SIE detection, we now characterize the synchronization properties that define these events.

3.5.1 Power Elevation

Power spectral elevation at SR harmonics during detected events was quantified using z-scores relative to session baselines. The fundamental frequency SR1 showed strongest elevation ($z = 4.76 \pm 2.95$), with systematic decrease at higher harmonics (Table 11).

Harmonic	Mean z	Median z	SD
SR1	4.76	3.88	2.95
SR1.5	3.95	3.18	3.08
SR2	3.91	3.16	3.01
SR2o	3.89	3.11	3.04
SR2.5	3.85	3.09	2.91
SR3	3.73	2.99	2.83
SR4	3.72	3.07	2.80
SR5	3.64	2.99	2.90
SR6	3.70	3.04	2.66

All harmonics showed substantial power elevation (all median $z > 2.7$), confirming multi-band spectral enhancement during SIE events.

3.5.2 Coherence and Phase-Locking

Magnitude-squared coherence (MSC) and phase-locking value (PLV) quantified inter-regional synchronization at each harmonic (Table 12).

Harmonic	Mean MSC	Mean PLV
SR1	0.552	0.689
SR1.5	0.532	0.672
SR2	0.538	0.681
SR2o	0.542	0.686
SR2.5	0.540	0.688
SR3	0.524	0.684
SR4	0.521	0.684
SR5	0.501	0.663
SR6	0.478	0.653

Both MSC and PLV showed gradients decreasing from SR1 to SR6, with SR1 exhibiting highest synchronization (MSC = 0.552, PLV = 0.689). Notably, 60.5% of events showed MSC > 0.5 and 81.5% showed PLV > 0.6 at the fundamental frequency, confirming robust whole-brain synchronization during SIE events consistent with communication through coherence (Fries, 2015a).

3.5.3 Cross-Frequency Coupling

Correlation analysis of z-scores across harmonics revealed strong cross-frequency coupling (Table 13).

Pair	Pearson r
SR1 vs SR3	0.74
SR3 vs SR5	0.77
SR5 vs SR6	0.83
SR1 vs SR6	0.53

MSC values showed even stronger coupling at higher harmonics (SR5–SR6: $r = 0.87$), suggesting coordinated synchronization across the harmonic stack during SIE events. This cross-frequency coupling in amplitude/coherence metrics contrasts with the independence observed in frequency values, indicating that SIEs involve coordinated power enhancement across independently-tuned oscillatory generators.

3.6 Event Quality Assessment

With the robustness of SIE detection established, we now examine what distinguishes high-quality from low-quality events.

3.6.1 SR-Score Distribution

The SR-Score (computed from sr1, sr3, sr5 with fixed weights) showed the following distribution across 1,366 events:

- Mean: 2.101 ± 1.879
- Median: 1.599
- Range: 0.060–30.152
- Quartiles: $Q1 = 1.032$, $Q2 = 1.599$, $Q3 = 2.570$

3.6.2 Harmonic Stack Index (HSI)

HSI, measuring the ratio of overtone power to fundamental power, showed a median of 0.54. This indicates that summed overtone power was approximately 54% of fundamental power during detected events—consistent with fundamental-dominated ignition, where the SR1 frequency drives the synchronized state while higher harmonics contribute secondary power.

3.6.3 Relationship Between Quality and φ -Precision

Correlation analysis revealed no significant relationship between SR-Score and φ^2 error magnitude ($r = +0.052$, $p = 0.10$). High-quality and low-quality events showed comparable φ -precision (high SR-Score: 4.27% error; low SR-Score: 4.17% error; $t = 0.58$, $p = 0.56$), indicating that

golden ratio organization is a fundamental property of SIE events rather than an artifact of event quality or signal-to-noise ratio.

3.6.4 Quality Determinants

The SR-Score showed strong correlations with individual synchronization measures (Table 20).

Table 20. SR-Score Correlations

Metric	Pearson r	p -value
z -score (SR1)	+0.697	$< 10^{-198}$
MSC (SR1)	+0.656	$< 10^{-168}$
PLV (SR1)	+0.451	$< 10^{-68}$
HSI (inverse)	-0.271	$< 10^{-23}$

Power elevation at SR1 was the strongest determinant of event quality ($r = 0.697$), followed by coherence ($r = 0.656$) and phase-locking ($r = 0.451$). This hierarchy suggests that amplitude increases at the fundamental frequency, supported by network-wide phase synchronization, are the primary characteristics distinguishing high-quality SIE events.

3.7 Temporal Dynamics and Network Organization

Beyond the static characterization of SIE properties, we examined the temporal evolution of ignition events to understand the dynamics underlying their formation.

3.7.1 Six-Phase Temporal Evolution

Individual SIE events exhibited a stereotyped temporal evolution characterized by six distinct phases (Figure 3). Analysis of Kuramoto order parameter $R(t)$ trajectories across all 1,366 events revealed a canonical ignition sequence:

1. **Phase 1 (Baseline):** Low-to-moderate global synchronization ($R \approx 0.4$ – 0.6) with characteristic fluctuations reflecting ongoing neural activity.
2. **Phase 2 (Coherence Rise):** Rapid increase in phase alignment preceding amplitude changes, with R typically rising from baseline to > 0.7 over 1–3 seconds. This coherence-first signature is consistent with transient large-scale integration (Varela et al., 2001) rather than local oscillatory bursts.
3. **Phase 3 (Plateau):** Brief stabilization (0.5–2 seconds) at elevated coherence levels ($R > 0.8$) immediately prior to amplitude ignition.
4. **Phase 4 (Ignition):** Coincident peak in both coherence and power, marked by simultaneous z -score elevation across multiple harmonics. The fundamental frequency SR1 showed the most pronounced elevation (mean $z = 4.57$, 97.8% of events exceeding 3σ).

5. **Phase 5 (Propagation):** Sustained high-coherence state (5–15 seconds) with gradual spread of synchronization across harmonic frequencies. Cross-frequency coupling metrics (bicoherence, PAC) showed maximal values during this phase.
6. **Phase 6 (Decay):** Gradual return to baseline over 3–8 seconds, with coherence typically declining before power—the inverse of the ignition sequence.

3.8 Emergent Frequency Properties

Beyond the basic characterization of SIE events, analysis of the measured harmonic frequencies revealed unexpected mathematical structure.

3.8.1 Golden Ratio (φ^n) Frequency Organization

Upon examining the nine harmonic frequencies detected across events (Table 4), a striking pattern emerged: the measured frequencies align closely with a geometric series $f(n) = f_0 \times \varphi^n$ where $\varphi = 1.618034$ (the golden ratio). Using an empirically-derived fundamental frequency $f_0 = 7.6$ Hz, measured frequencies matched theoretical predictions across integer, half-integer, and quarter-integer values of n .

Harmonic number (n) assignment methodology: The assignment of specific n values to detected peaks was *not* a post-hoc optimization but followed a constrained procedure: (1) the fundamental SR1 was assigned $n = 0$ by definition, anchoring the scale; (2) integer n values (1, 2, 3) were assigned to peaks nearest the canonical Schumann Resonance modes, preserving the physical correspondence to electromagnetic cavity modes; (3) half-integer and quarter-integer values were introduced only where robust spectral peaks existed at intermediate frequencies that did not match integer predictions.

Four of nine harmonics showed prediction errors below 1%, with mean absolute error across all harmonics of 1.42% and median error of 1.18%. The integer harmonics ($\varphi^0, \varphi^1, \varphi^2, \varphi^3$) showed systematically better fit compared to half-integer harmonics, though this difference did not reach statistical significance. The quarter-integer harmonic SR2o ($\varphi^{1.25}$) showed excellent precision (0.76% error), supporting the presence of fine-grained φ^n structure.

Optimal Fundamental Frequency. To identify the f_0 value providing best overall fit, we computed mean absolute prediction error across all harmonics for candidate f_0 values spanning 7.0–8.0 Hz (Figure 2A). Optimization yielded optimal $f_0 = 7.58$ Hz (minimizing squared error), 7.54 Hz (minimizing absolute error), and 7.58 Hz (N-weighted squared error).

Table 5. Fundamental Frequency Comparison

f_0 Value (Hz)	Source	Mean Error	Max Error
7.494	Theoretical ($c/2\pi r$)	1.57%	4.60%
7.540	Optimal (min absolute)	1.38%	3.96%
7.580	Optimal (min squared)	1.39%	3.30%
7.600	Empirical (rounded)	1.45%	3.14%
7.630	Measured SR1 mean	1.49%	3.13%
7.830	Schumann Resonance fundamental	3.28%	5.61%

The empirical $f_0 = 7.6$ Hz provides near-optimal fit (1.45% mean error) while maintaining interpretability. Critically, 7.6 Hz also corresponds to the empirical center of measured Schumann Resonance fluctuations: long-term monitoring data from the Space Observing System facility in Tomsk, Russia shows the SR fundamental varies around 7.6 ± 0.6 Hz across hours, days, and seasons (Nickolaenko and Hayakawa, 2014). Thus $f_0 = 7.6$ Hz represents both the empirically optimal fit to our EEG data and the measured center of SR fluctuation range. The canonical 7.83 Hz represents the modal peak of SR power spectral density, whereas 7.6 Hz represents the center of the frequency fluctuation window; our data suggest neural frequencies may track the latter rather than the former.

Golden Ratio Precision in Harmonic Ratios. The φ^n architecture predicts specific ratios between harmonics: $\text{SR3}/\text{SR1} = \varphi^2$, $\text{SR5}/\text{SR1} = \varphi^3$, $\text{SR5}/\text{SR3} = \varphi$, and $\text{SR6}/\text{SR4} = \varphi$. These ratios are independent of f_0 choice and provide the most stringent test of golden ratio organization (Table 6).

Ratio	Predicted	Measured	Error (%)	95% CI
$\text{SR3}/\text{SR1} = \varphi^2$	2.6180	2.6223 ± 0.134	+0.16	[2.614, 2.630]
$\text{SR5}/\text{SR1} = \varphi^3$	4.2361	4.2766 ± 0.249	+0.96	[4.262, 4.291]
$\text{SR5}/\text{SR3} = \varphi$	1.6180	1.6309 ± 0.078	+0.79	[1.627, 1.635]
$\text{SR6}/\text{SR4} = \varphi$	1.6180	1.6094 ± 0.100	-0.53	[1.604, 1.615]

Mean absolute ratio error: 0.61% All four ratios deviated less than 1% from predicted φ^n values, with mean absolute error of 0.61%. Bootstrap analysis ($N = 10,000$ iterations) confirmed that the 95% confidence interval for $\text{SR3}/\text{SR1}$ (2.614–2.630) includes the predicted φ^2 value (2.618). However, the CI for $\text{SR5}/\text{SR3}$ (1.627–1.635) excludes the exact φ value (1.618), indicating a small but consistent positive deviation of approximately 0.8%.

Note on error metrics: This 0.61% represents the *population-level* ratio error, computed using mean frequencies across all events (e.g., mean SR3 / mean SR1). Individual event-level precision is lower: when each event’s φ -error is computed separately and then averaged across all 1,366 events, the mean composite φ -error is 4.37%. This ~ 7 -fold difference is expected from the Central Limit Theorem: averaging $\sim 1,300$ independent measurements reduces standard error by a factor of $\sqrt{1300} \approx 36$, yielding much higher precision for population means than for individual events. Both metrics are valid but measure different quantities: population-level precision (0.61%) reflects the central tendency of the φ^n architecture, while event-level precision (4.37%) captures within-event variability.

3.8.2 The Independence-Convergence Paradox

A critical question concerns whether the three primary harmonics (SR1, SR3, SR5) vary together proportionally or independently. If frequencies were proportionally coupled (i.e., $\text{SR3} \approx k \times \text{SR1}$), we would expect strong positive correlations between harmonic frequencies across events. Instead, pairwise Pearson correlations revealed complete independence (Table 7).

Pair	Pearson r	p -value	Interpretation
SR1 vs SR3	+0.030	0.33	No correlation
SR1 vs SR5	−0.002	0.94	No correlation
SR3 vs SR5	−0.020	0.47	No correlation

All correlations were statistically non-significant (all $p > 0.3$), with r values indistinguishable from zero. This demonstrates that harmonic frequencies vary completely independently across events—an event with high SR1 frequency is no more likely to have high SR3 or SR5 frequencies than expected by chance.

Compensatory Error Mechanism. Despite frequency independence, harmonic ratios maintain remarkable precision ($< 1\%$ error). This paradox resolves through analysis of ratio errors, which revealed a significant compensatory correlation (Table 8).

Error Pair	Pearson r	p -value
SR3/SR1 error vs SR5/SR3 error	−0.314	$< 10^{-24}$
SR3/SR1 error vs SR5/SR1 error	+0.620	$< 10^{-100}$
SR5/SR1 error vs SR5/SR3 error	+0.550	$< 10^{-80}$

The critical finding is the strong negative correlation ($r = -0.314$, $p < 10^{-24}$) between SR3/SR1 error and SR5/SR3 error: when SR3/SR1 deviates above φ^2 , SR5/SR3 compensatorily deviates below φ , and vice versa. This compensatory mechanism maintains overall φ^n precision despite independent frequency variation, suggesting intrinsic mathematical constraints rather than coupled oscillatory generators.

Population-Level φ^n Encoding. To test whether ratio precision emerges from event-level harmonic coordination or population-level frequency distributions, we performed a shuffled bootstrap analysis. SR1, SR3, and SR5 frequencies were independently permuted across events ($N = 10,000$ iterations), destroying within-event pairings while preserving marginal distributions.

Shuffled Bootstrap Results:

- Observed mean composite φ -error: 4.37%
- Shuffled null mean error: $4.36\% \pm 0.05\%$
- p -value: 0.60 (not significant)
- Effect size (Cohen’s d): −0.27

The non-significant p -value indicates that shuffling harmonic frequencies does not degrade ratio precision, implying that φ^n relationships are encoded in the marginal frequency distributions themselves rather than through event-level coordination. The brain independently constrains each harmonic’s frequency distribution to centers that satisfy φ^n relationships at the population level.

3.8.3 Frequency Stability Hierarchy

Coefficient of variation (CV) analysis across harmonics revealed a systematic pattern of frequency stability. Among the primary harmonics (SR1–SR5), SR5 (β/γ boundary; $CV = 1.79\%$) showed the lowest variability, followed by SR3 (β boundary; $CV = 2.02\%$). The fundamental frequency SR1 showed highest variability ($CV = 3.18\%$). This pattern of decreasing CV from SR1 to SR5 suggests that higher harmonics show greater frequency stability than the fundamental, contradicting simple bottom-up models where fundamental frequency precision should propagate to higher harmonics. The beta-range harmonics (SR3, SR5) may serve as more stable reference frequencies than the theta-range fundamental (SR1), consistent with the two-subsystem model: a theta-centered system with moderate stability, and a tightly coupled beta-gamma complex with higher precision.

3.8.4 Beat Frequency Architecture

Inter-harmonic beat frequencies revealed additional structure consistent with φ^n architecture. Both SR3–SR1 and SR5–SR3 beat frequencies approximate 12.4 Hz—the φ^1 frequency (α/β boundary)—suggesting nested oscillatory structure where inter-harmonic differences themselves follow φ^n relationships. Specifically, SR3 – SR1 measured 12.38 ± 0.48 Hz (predicted: 12.13 Hz) and SR5 – SR3 measured 12.49 ± 0.70 Hz (predicted: 12.13 Hz).

4 Discussion

4.1 Summary of Principal Findings

The present study provides the first systematic characterization of Schumann Ignition Events (SIEs)—transient, high-coherence neural states exhibiting precise frequency organization at golden ratio (ϕ^n) scaled harmonics. Analysis of 1,366 events across 91 participants, five cognitive contexts, and three EEG recording devices revealed several principal findings:

1. **Reproducible neural phenomenon:** SIEs are characterized by brief (median 20 seconds) episodes of network-wide synchronization, with 81.5% of events showing phase-locking values exceeding 0.6 and 60.5% showing magnitude squared coherence exceeding 0.5 at the fundamental frequency.
2. **Golden ratio frequency organization:** Harmonic frequencies spanning 7–40 Hz are organized according to powers of the golden ratio ($\phi = 1.618$), with measured frequencies deviating less than 1.5% from theoretical predictions $f(n) = f_0 \times \phi^n$ for an empirically-derived fundamental $f_0 = 7.6$ Hz.
3. **Independence-convergence paradox:** Individual harmonic frequencies vary completely independently across events (all pairwise correlations $|r| < 0.03$), yet their population-level ratios maintain remarkable precision ($<1\%$ error from ϕ^n predictions; event-level precision $\sim 4.4\%$). This paradox suggests that golden ratio relationships are encoded in the marginal distributions of each harmonic rather than through event-level coordination.

4. **Stereotyped temporal dynamics:** SIEs exhibit a six-phase temporal evolution in which coherence elevation precedes amplitude increases by 2–3 seconds, suggesting network-level coordination rather than local oscillatory bursts.
5. **Cross-context and cross-device stability:** The ϕ^n frequency architecture is preserved across meditation, cognitive flow, non-flow arithmetic, gaming, and passive visual perception tasks, and across consumer EEG devices with substantially different electrode configurations, indicating a fundamental organizational property rather than task-specific artifact. Higher harmonics (SR5) showed modest device-related frequency shifts.

4.2 Relationship to Transient Synchronization Phenomena

The temporal dynamics of SIEs share features with previously described transient synchronization phenomena. The coherence-first signature—phase alignment preceding amplitude elevation—is consistent with the “ignition” dynamics described in Global Neuronal Workspace theory (Dehaene and Changeux, 2011; Mashour et al., 2020). In this framework, conscious access is characterized by a late, large-scale ignition event involving prefrontal-parietal networks with sudden gamma-band enhancement and long-distance phase synchronization (Gaillard et al., 2009).

Similarly, the six-phase temporal evolution parallels the “metastable” dynamics observed in large-scale brain networks (Deco and Kringelbach, 2017; Tognoli and Kelso, 2014). Metastability refers to a dynamical regime intermediate between complete synchronization and complete independence, where periods of high synchronization alternate with desynchronized states in a continuous flow of transient coordination patterns. SIEs may represent particularly robust instances of such metastable coordination.

However, SIEs differ from previously characterized transient synchronization in several respects. First, the frequency specificity is unprecedented: rather than broadband synchronization or synchronization within conventional frequency bands, SIEs involve coordinated activity at nine discrete frequencies spanning theta through gamma that satisfy precise mathematical relationships. Second, the duration of 10–30 seconds exceeds typical gamma bursts (100–500 ms) while remaining shorter than sustained task-related oscillatory changes. Third, the cross-frequency coupling architecture—with a theta-anchored subsystem (SR1) showing moderate coupling to higher harmonics and a tightly coupled beta-gamma complex (SR3–SR6)—suggests hierarchical organization not captured by single-frequency synchronization accounts.

The discovery that SR-frequency organization exists continuously at baseline, with SIEs representing transient amplification rather than de novo generation, is consistent with the intrinsic ignition framework proposing that spontaneous brain dynamics constantly generate transient coordination events (Deco and Kringelbach, 2017). This “continuous-plus-amplification” model has implications for understanding how the brain rapidly recruits coordinated network activity without requiring complete reorganization of oscillatory structure.

4.3 The Independence-Convergence Paradox: A Novel Organizational Principle

The most theoretically significant finding is the independence-convergence paradox: harmonic frequencies (SR1, SR3, SR5) vary completely independently across events (all correlations indistinguishable from zero), yet their ratios maintain less than 1% deviation from golden ratio predictions. This paradox resolves through analysis showing a compensatory error mechanism ($r = -0.314$ between SR3/SR1 error and SR5/SR3 error) and shuffled bootstrap validation demonstrating that ϕ^n precision is preserved when frequencies are randomly permuted across events.

These findings indicate that golden ratio relationships are encoded in the population-level marginal distributions of each harmonic frequency, not through event-level coordination between oscillatory generators. Each harmonic appears to be independently constrained to a distribution whose mean satisfies ϕ^n relationships relative to other harmonics. This organizational principle differs fundamentally from both simple proportional coupling (where harmonic frequencies would covary) and independent oscillators (where ratio precision would degrade with frequency independence).

What biological mechanism could produce such population-level coordination among independently varying frequencies? One possibility is that each frequency band is constrained by local biophysical properties—membrane time constants, network architecture, receptor kinetics—that independently converge on ϕ^n values. The GABA-A receptor decay time constant (~ 25 ms for perisomatic basket cell synapses) directly determines gamma oscillation frequency through Pyramidal-Interneuron Network Gamma (PING) mechanisms (Bartos et al., 2007; Brunel and Wang, 2003; Wang, 2010). Parvalbumin-positive fast-spiking interneurons, which constitute the primary cellular substrate for gamma generation (Cardin et al., 2009; Sohal et al., 2009), show intrinsic resonance at approximately 40 Hz. If these timescales have been evolutionarily tuned to produce compatible frequency relationships—perhaps to optimize cross-frequency information transfer—independent variation within each system could still satisfy population-level ϕ^n constraints.

Alternatively, developmental processes might establish golden ratio relationships during circuit formation, with mature oscillatory generators retaining the mathematical structure while acquiring independent variability. Evidence that individual alpha frequency is highly heritable ($h^2 = 0.81$; Posthuma et al., 2001; Smit et al., 2006) and stable across the lifespan despite substantial within-session variability (Grandy et al., 2013) supports the notion of genetically determined frequency “set points” that could encode ϕ^n architecture at the population level.

4.4 Golden Ratio Organization and EEG Frequency Band Structure

The ϕ^n frequency architecture provides a principled basis for understanding the organization of canonical EEG frequency bands (Klimesch, 1999a). Traditional band definitions (delta 1–4 Hz, theta 4–8 Hz, alpha 8–13 Hz, beta 13–30 Hz, gamma >30 Hz) have been established empirically over decades of research, but the boundaries between bands have remained somewhat arbitrary, with different researchers using different cutoffs and no underlying principle explaining why these particular frequency ranges should be functionally distinct.

The present findings suggest that EEG band boundaries may correspond to integer powers of ϕ (boundary frequencies), while band centers correspond to half-integer powers (attractor frequencies). Specifically: the theta-alpha boundary (~ 8 Hz) aligns with ϕ^0 (7.6 Hz); the alpha-beta boundary (~ 12 Hz) aligns with ϕ^1 (12.3 Hz); the beta boundary (~ 20 Hz) aligns with ϕ^2 (19.9 Hz); and the beta-gamma boundary (~ 32 Hz) aligns with ϕ^3 (32.2 Hz). The canonical alpha “peak” (~ 10 Hz) corresponds to the $\phi^{0.5}$ attractor (9.7 Hz); the sensorimotor rhythm (~ 15 Hz) corresponds to $\phi^{1.5}$ (15.6 Hz); and the 40 Hz gamma attractor corresponds to $\phi^{3.5}$ (40.9 Hz).

This framework aligns with large-scale spectral parameterization data showing that FOOOF-detected EEG peaks cluster at half-integer ϕ^n positions and are depleted at integer ϕ^n positions (Donoghue et al., 2020), consistent with the attractor-boundary distinction. The framework also aligns with (buzsaki2004neural) observation that mean brain oscillation frequencies form a linear progression on a natural logarithmic scale—the golden ratio ($\phi \approx 1.618$) is close to Euler’s number ($e \approx 2.718$), and both predict approximately geometric frequency spacing.

The functional implications of ϕ^n organization extend to cross-frequency coupling. Phase-amplitude coupling between theta and gamma, alpha and gamma, and delta-theta and alpha-beta creates nested oscillatory hierarchies that transfer information across spatial and temporal scales (Canolty and Knight, 2010a; Jensen and Colgin, 2007). If these coupled frequencies satisfy ϕ^n relationships, non-integer ratios would minimize harmonic interference while maintaining mathematical precision—potentially optimizing information transfer across the frequency hierarchy (Bonnefond et al., 2017).

Intriguingly, the ϕ^n architecture exhibits self-similar structure in beat frequencies. Inter-harmonic beat frequencies ($\text{SR3} - \text{SR1} = 12.38 \pm 0.48$ Hz; $\text{SR5} - \text{SR3} = 12.49 \pm 0.70$ Hz) both approximate 12.4 Hz—the ϕ^1 frequency corresponding to the alpha-beta boundary. This pattern suggests nested oscillatory structure where not only the harmonic frequencies themselves, but also their differences, follow ϕ^n relationships. Such recursive mathematical organization could provide multiple channels for cross-frequency information transfer: direct phase-amplitude coupling between harmonics, and amplitude-amplitude interactions at beat frequencies. The convergence of beat frequencies on ϕ^1 may also explain why the alpha-beta transition zone (~ 12 Hz) is functionally significant across diverse cognitive tasks—it represents not just a boundary frequency but also the primary beat frequency generated by theta-beta and beta-gamma interactions.

4.5 Two Synchronization Subsystems

Analysis of cross-frequency amplitude correlations revealed a distinctive architecture suggesting two partially dissociable synchronization subsystems. The fundamental frequency SR1 (7.6 Hz) showed moderate coupling to higher harmonics ($r = 0.32\text{--}0.54$), while the beta-gamma harmonics SR3–SR6 (20–40 Hz) formed a tightly coupled complex (all pairwise $r > 0.6$). Importantly, these correlations reflect coordinated amplitude/power changes across harmonics—not frequency correlations, which remain near zero (see Independence-Convergence Paradox above).

This architecture aligns with established distinctions between theta-based and beta/gamma-based coordination mechanisms. Theta oscillations are associated with long-range hippocampal-

cortical communication (Colgin et al., 2009; Hyman et al., 2005), working memory maintenance, and temporal organization of sequential information (Sauseng et al., 2009). Beta oscillations, in contrast, are associated with maintaining the “status quo” (Engel and Fries, 2010; Spitzer and Haegens, 2017), sensorimotor integration, and long-range cortical synchronization where conduction delays challenge gamma coherence. The relatively weak coupling between SR1 and SR6 ($r = 0.32$) despite strong coupling within each subsystem suggests that theta-anchored and beta-gamma coordination mechanisms may be recruited somewhat independently during SIE ignition, with full integration occurring only at the population level through the ϕ^n constraint.

Having characterized the frequency organization and network architecture of SIEs, we now address the terminology and its relationship to geophysical phenomena.

4.6 Relationship to Schumann Resonance

The term “Schumann Ignition Events” references the correspondence between observed neural frequencies and the Schumann Resonances—extremely low-frequency electromagnetic phenomena arising from the Earth-ionosphere cavity, with a fundamental mode at approximately 7.83 Hz and higher harmonics at approximately 14.3, 20.8, 27.3, and 33.8 Hz (Nickolaenko and Hayakawa, 2014). Several aspects of this correspondence warrant careful interpretation.

First, the relationship is one of frequency correspondence, not demonstrated coupling. The measured Schumann Resonance fundamental (7.83 Hz) provides the poorest fit to observed neural frequencies (3.28% mean error), while the theoretical electromagnetic prediction based on Earth’s circumference ($c/2\pi r = 7.494$ Hz) and the empirically-derived optimal f_0 (7.54–7.60 Hz) provide better fits (1.38–1.45% error). This suggests that neural ϕ^n architecture is not precisely tuned to measured SR frequencies, but rather to values approximately 3% below the SR fundamental.

Second, without concurrent magnetometer recording of actual Schumann Resonance field strength, we cannot assess whether SIE occurrence correlates with SR activity or whether any causal relationship exists between planetary electromagnetic resonances and neural oscillations. Prior studies reporting correlations between SR intensity and human physiological variables have been limited by methodological concerns (Cherry, 2002; Price et al., 2016; Saroka et al., 2016), and proposed mechanisms for SR-brain coupling remain speculative given the extremely weak field strengths ($\sim$ picoTesla) of SR signals.

Third, an alternative interpretation is that the frequency correspondence reflects convergent constraints rather than direct coupling. Both the Earth-ionosphere cavity (electromagnetic resonance) and the brain (neural oscillations) may independently optimize to similar frequencies due to shared physical principles—characteristic length scales, propagation velocities, and resonance conditions. The SR fundamental frequency is determined by $c/2\pi r$ where c is the speed of light and r is Earth’s radius; interestingly, the theoretical neural f_0 providing best ϕ^n fit (7.54 Hz) is close to this electromagnetic prediction.

We therefore emphasize that “Schumann” in our terminology refers to the frequency range correspondence and the historical observation that human EEG and SR occupy overlapping spectral bands, not to proven electromagnetic coupling. Whether this correspondence reflects evolutionary adaptation to the planetary electromagnetic environment, coincidental convergence

of physical constraints, or artifact of selection bias in EEG band definitions remains an open question requiring concurrent SR-EEG recording for resolution.

Regardless of the electromagnetic question, the neural phenomena themselves warrant functional interpretation.

4.7 Functional Significance of SIEs

What functional role might SIEs serve? Several lines of evidence suggest involvement in states of heightened cognitive integration. First, SIE detection rates varied substantially across cognitive contexts, with cognitive flow/non-flow tasks showing the highest rate (9.7 events/session) followed by meditation (7.6/session), gaming (~ 1.1 /session), and passive visual perception (~ 1.1 /session). While recording duration differences may contribute to this pattern, the elevated rates during flow states and meditation—both characterized by absorbed attention and reduced mind-wandering—suggest a relationship between SIEs and focused cognitive engagement.

Second, the temporal dynamics of SIEs—coherence preceding amplitude, fundamental dominance preceding ignition, and frontal-to-posterior information flow during peak synchronization—are consistent with top-down coordination mechanisms associated with attentional control and conscious access. The observation that high-quality events show dramatically stronger fundamental dominance (HSI 75% lower, indicating the fundamental SR1 frequency dominates over overtones) and higher coherence (82% higher) than low-quality events suggests that SR1-led frequency organization is functionally meaningful rather than epiphenomenal.

Third, the multi-band nature of SIEs—involving coordinated activity across nine harmonics spanning theta through gamma—distinguishes them from single-frequency phenomena and suggests involvement in multi-scale information integration. The hierarchical frequency organization described in predictive coding frameworks (Bastos et al., 2012; Friston and Kiebel, 2009), where gamma carries feedforward prediction errors and alpha/beta carry feedback predictions, may require precisely-tuned frequency relationships to function optimally. SIEs may represent moments when this hierarchical architecture achieves maximal coordination.

Fourth, nonlinear cross-frequency coupling metrics reveal coordinated harmonic interactions during SIE events. Bicoherence analysis—which quantifies quadratic phase coupling between frequency triplets—showed multiple harmonic triads simultaneously surging to values exceeding 0.8 during the coherence/plateau phases, compared to baseline levels of 0.2–0.4. This synchronized increase across different frequency combinations indicates that nonlinear harmonic generation involves the entire ϕ^n stack through a common generative mechanism rather than independent oscillator coupling. Notably, bicoherence peaks slightly before maximal power elevation, suggesting that nonlinear coupling may drive harmonic power amplification rather than emerge as a consequence of it. Similarly, theta-gamma phase-amplitude coupling (PAC), measured by mean vector length, approximately triples during ignition (from ~ 0.05 to ~ 0.20), indicating substantial enhancement of hierarchical frequency organization consistent with nested oscillatory coding (J. E. Lisman and Jensen, 2013).

Fifth, examination of state-space dynamics reveals that SIE events occupy a qualitatively distinct neural regime. When plotting instantaneous coherence (Kuramoto order parameter)

against harmonic power, baseline activity clusters at low power with moderate coherence ($R \sim 0.5\text{--}0.7$), while ignition trajectories extend to high power while maintaining or increasing coherence ($R \sim 0.7\text{--}0.95$), forming a characteristic “lasso” pattern. The clear non-overlap between baseline and ignition clusters demonstrates that SIEs are not simply extreme values of a continuous distribution but represent a separate attractor state in neural dynamics.

Sixth, the finding that power elevation at SR1 is the strongest determinant of event quality ($r = 0.697$), followed closely by coherence ($r = 0.656$), suggests that amplitude increases at the fundamental frequency, supported by network-wide phase synchronization, are the primary characteristics distinguishing high-quality SIE events. The strong but secondary role of coherence aligns with communication-through-coherence frameworks (Fries, 2015a), where phase alignment between oscillatory populations gates the information flow that amplified signals carry.

Finally, the Harmonic Stack Index (HSI) findings—where fundamental dominance (decreasing HSI) precedes ignition by ~ 3 seconds—suggest that SR1-led frequency organization may be a prerequisite for, rather than consequence of, ignition. The brain appears to first establish fundamental dominance over overtones before power can increase, implying that the SR1 frequency must lead the harmonic stack to support the subsequent synchronization cascade.

However, the functional significance of SIEs remains speculative without direct manipulation or behavioral correlates. Future studies should examine whether SIE occurrence predicts task performance, whether SIE frequency precision correlates with cognitive measures, and whether entrainment at ϕ^n frequencies enhances cognitive function relative to entrainment at non- ϕ^n frequencies.

4.8 Methodological Considerations

Several methodological factors warrant consideration in interpreting these findings.

The use of consumer-grade EEG devices (Emotiv EPOC X, Muse, Insight) with limited channel counts (4–14 channels) and lower signal-to-noise ratio compared to research-grade systems raises questions about spatial resolution and artifact contamination. However, these devices have been validated for research applications (Badcock et al., 2015; Krigolson et al., 2017).

Several observations argue against device-specific artifacts. First, cross-device frequency convergence was excellent, with means within 0.3% of the grand mean for SR1 and SR3. Second, phase randomization—which preserves spectral content while destroying phase relationships—dramatically reduced synchronization metrics, indicating that the coherence signatures depend on genuine phase relationships rather than spectral artifacts. Third, the consistency of findings across devices with different electrode technologies (dry, semi-dry polymer, saline-soaked felt) and spatial coverage (frontal-temporal to comprehensive scalp) suggests biological rather than technical origins.

However, SR5 frequencies showed a significant device effect ($F = 5.74$, $p = 0.003$), with Muse devices detecting systematically higher frequencies than Emotiv devices (~ 0.35 Hz difference). This may reflect differences in anti-aliasing filters or spectral sensitivity at higher frequencies. While this does not invalidate the ϕ^n architecture—which is defined by frequency ratios rather than absolute frequencies—it indicates that absolute frequency claims for higher harmonics should specify the recording device.

The discovery phase relied on longitudinal recordings from a single experienced meditation practitioner, raising concerns about generalizability and potential overfitting of detection parameters. However, the validation phase demonstrated comparable SIE characteristics in 58 additional participants performing cognitive tasks unrelated to meditation, with no significant frequency differences between datasets.

The seven-stage detection pipeline involves multiple parameter choices (z-threshold, window duration, merging criteria, quality thresholds) that could influence detected event characteristics. Sensitivity analyses varying detection thresholds would strengthen confidence in the robustness of findings, though the large effect sizes observed ($d > 1.4$ for key comparisons) suggest that results would survive reasonable parameter variations.

4.9 Limitations

Several limitations constrain interpretation of the present findings. First, without concurrent magnetometer recording of Schumann Resonance field strength, we cannot assess whether SIE occurrence correlates with actual SR activity or whether any causal relationship exists. Claims about brain-environment electromagnetic coupling remain entirely speculative.

Second, the consumer EEG devices used provide limited spatial resolution. While the transfer entropy analysis suggested frontal hub topology during ignition, this conclusion is constrained by sparse electrode coverage and potential volume conduction effects. High-density EEG or source localization approaches would provide stronger evidence regarding the spatial organization of SIE generators.

Third, the present study is descriptive rather than mechanistic. We document the existence and characteristics of SIEs but cannot explain why neural oscillations would organize according to golden ratio mathematics. The mechanism generating ϕ^n architecture—whether evolutionary adaptation, developmental constraint, or emergent property of coupled oscillators—remains unknown.

Fourth, we did not examine behavioral correlates of SIE occurrence within events. Whether SIEs are associated with specific cognitive states, whether they predict task performance, and whether they are accessible to awareness remain open questions.

Fifth, the single-subject discovery dataset, while subsequently validated in independent samples, limits confidence that detection parameters generalize optimally across populations. Individual differences in baseline oscillatory dynamics, age effects on peak frequencies, and clinical variation could all influence SIE detection and characteristics.

4.10 Future Directions

Several directions would advance understanding of SIE phenomena. First, concurrent recording of EEG and Schumann Resonance field strength using magnetometers would enable direct assessment of brain-environment coupling hypotheses. If SIE occurrence correlates with SR field intensity or phase, this would support active coupling; if not, the frequency correspondence may reflect convergent constraints rather than electromagnetic interaction.

Second, high-density EEG with source localization would clarify the spatial organization of SIE generators. The present findings suggest frontal hub topology, but confirmation requires

better spatial resolution than consumer devices provide.

Third, entrainment studies using transcranial alternating current stimulation (tACS; Hermann et al., 2013; Pogosyan et al., 2009) or sensory flicker (Galambos et al., 1981) at ϕ^n versus non- ϕ^n frequencies would test whether the brain preferentially entrains to golden ratio frequencies. Prior work showing robust aftereffects of alpha-frequency tACS (Kasten et al., 2016; Zaehle et al., 2010) suggests that frequency-specific entrainment can produce lasting neural effects. If ϕ^n frequencies produce stronger entrainment or superior cognitive effects, this would support functional significance of the architecture.

Fourth, examination of SIEs in clinical populations—particularly those with altered oscillatory dynamics such as schizophrenia (gamma abnormalities; Kwon et al., 1999; Uhlhaas and Singer, 2010), Parkinson’s disease (beta abnormalities; Brown, 2007; Little and Brown, 2013), or ADHD (theta abnormalities; Barry et al., 2003)—could reveal whether ϕ^n organization is disrupted in pathological states and whether restoration correlates with clinical improvement.

Fifth, cross-species validation would test whether ϕ^n architecture is evolutionarily conserved. Preliminary evidence suggests similar frequency organization across mammalian species despite 17,000-fold variation in brain volume (Buzsáki et al., 2013); remarkably, even cephalopods—separated from vertebrates by 550 million years of evolution—show spindle-like 12–18 Hz oscillations during sleep (Medendorp et al., 2024), suggesting these frequency bands represent convergent evolutionary solutions. Systematic examination of ϕ^n relationships across species would clarify whether this reflects fundamental biophysical constraints or convergent evolutionary optimization.

4.11 Conclusions

This study makes three principal contributions to our understanding of neural oscillatory organization.

First, we establish SIEs as a reproducible neural phenomenon. Across 1,366 events, 91 participants, five cognitive contexts, and three consumer EEG devices, Schumann Ignition Events showed consistent spectral, temporal, and network characteristics. The convergence of detection across devices with different electrode technologies and spatial coverage (4–14 channels) argues for biological rather than technical origins. The convergence of detection across devices with different electrode technologies and spatial coverage (4–14 channels) argues for biological rather than technical origins. The coherence-first temporal signature, frontal hub topology during ignition, and distinct state-space trajectories suggest that SIEs represent a qualitatively distinct neural regime rather than extreme values of continuous oscillatory dynamics.

Second, we identify the independence-convergence paradox as a novel organizational principle. Harmonic frequencies vary completely independently across events (all pairwise $|r| < 0.03$), yet their population-level ratios maintain $<1\%$ deviation from golden ratio predictions. This paradox cannot be explained by coupled oscillators (which would show frequency covariation) or by independent rhythms (which would degrade ratio precision). Instead, it implies that each oscillatory generator is independently constrained to a frequency distribution whose mean satisfies ϕ^n relationships—a form of distributed coordination that preserves mathematical structure without requiring direct coupling. The compensatory error mechanism ($r = -0.314$ between successive ratio errors) further suggests an intrinsic constraint maintaining ϕ^n architecture.

Third, we propose a unified ϕ^n framework for EEG frequency organization. The observed harmonics map systematically onto canonical EEG bands: integer ϕ^n values correspond to band boundaries (theta-alpha at ϕ^0 , alpha-beta at ϕ^1 , beta-gamma at ϕ^3), while half-integer values correspond to band centers or “attractors” (alpha peak at $\phi^{0.5}$, gamma at $\phi^{3.5}$). The self-similar beat frequency structure—where inter-harmonic differences also converge on ϕ^n values—suggests recursive mathematical organization that may optimize cross-frequency information transfer.

The correspondence between SIE frequencies and Earth’s Schumann Resonances remains an open question. The neural fundamental (7.6 Hz) lies closer to the theoretical electromagnetic prediction ($c/2\pi r = 7.49$ Hz) than to the measured SR peak (7.83 Hz), suggesting that any relationship may reflect convergent physical constraints rather than direct electromagnetic coupling. Resolution requires concurrent EEG and magnetometer recording.

What the present findings establish is that transient, high-coherence neural states exhibit exceptional mathematical precision in their frequency organization—precision that emerges at the population level despite complete independence at the event level. This distributed encoding of golden ratio relationships across independently varying oscillators represents a novel form of neural coordination warranting further mechanistic investigation.

References

- Albuquerque, I., Tiwari, A., Parent, M., Cassani, R., Gagnon, J.-F., Lafond, D., Tremblay, S., & Falk, T. H. (2024). PhySF: A physiological dataset of flow and cognitive load states. *Scientific Data*, 11, 457. <https://doi.org/10.1038/s41597-024-03295-1>
- Albuquerque, I., Monteiro, J., Tiwari, A., & Falk, T. H. (2023). A multimodal dataset for player engagement analysis in video games. *Computers in Biology and Medicine*, 157, 106730. <https://doi.org/10.1016/j.compbiomed.2023.106730>
- Allen, J. B. (1977). Short term spectral analysis, synthesis, and modification by discrete Fourier transform. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 25(3), 235–238. <https://doi.org/10.1109/TASSP.1977.1162950>
- Babayev, E. S., & Allahverdiyeva, A. A. (2007). Effects of geomagnetic activity variations on the physiological and psychological state of functionally healthy humans: Some results of Azerbaijani studies. *Advances in Space Research*, 40(12), 1941–1951. <https://doi.org/10.1016/j.asr.2007.02.099>
- Badcock, N. A., Mousikou, P., Mahajan, Y., de Lissa, P., Thie, J., & McArthur, G. (2015). Validation of the Emotiv EPOC EEG gaming system for measuring research quality auditory ERPs [Validation of Emotiv EPOC for research applications]. *PeerJ*, 3, e907. <https://doi.org/10.7717/peerj.907>
- Barry, R. J., Clarke, A. R., & Johnstone, S. J. (2003). A review of electrophysiology in attention-deficit/hyperactivity disorder: I. Qualitative and quantitative electroencephalography [Theta/beta ratio abnormalities in ADHD]. *Clinical Neurophysiology*, 114(2), 171–183. [https://doi.org/10.1016/S1388-2457\(02\)00362-0](https://doi.org/10.1016/S1388-2457(02)00362-0)

- Bartos, M., Vida, I., & Jonas, P. (2007). Synaptic mechanisms of synchronized gamma oscillations in inhibitory interneuron networks [GABA-A kinetics and gamma oscillation generation]. *Nature Reviews Neuroscience*, 8(1), 45–56. <https://doi.org/10.1038/nrn2044>
- Bastos, A. M., Usrey, W. M., Adams, R. A., Mangun, G. R., Fries, P., & Friston, K. J. (2012). Canonical microcircuits for predictive coding [Gamma feedforward, alpha/beta feedback in predictive coding]. *Neuron*, 76(4), 695–711. <https://doi.org/10.1016/j.neuron.2012.10.038>
- Bonnefond, M., Kastner, S., & Jensen, O. (2017). Communication between brain areas based on nested oscillations [Unification of communication through coherence and gating by inhibition]. *eNeuro*, 4(2), ENEURO.0153–16.2017. <https://doi.org/10.1523/ENEURO.0153-16.2017>
- Brown, P. (2007). Abnormal oscillatory synchronisation in the motor system leads to impaired movement [Pathological beta in Parkinson’s disease]. *Current Opinion in Neurobiology*, 17(6), 656–664. <https://doi.org/10.1016/j.conb.2007.12.001>
- Brunel, N., & Wang, X.-J. (2003). What determines the frequency of fast network oscillations with irregular neural discharges? I. Synaptic dynamics and excitation-inhibition balance [PING/ING models of gamma generation]. *Journal of Neurophysiology*, 90(1), 415–430. <https://doi.org/10.1152/jn.01095.2002>
- Butterworth, S. (1930). On the theory of filter amplifiers. *Wireless Engineer*, 7(6), 536–541.
- Buzsáki, G. (2015). Hippocampal sharp wave-ripple: A cognitive biomarker for episodic memory and planning. *Hippocampus*, 25(10), 1073–1188. <https://doi.org/10.1002/hipo.22488>
- Buzsáki, G., & Draguhn, A. (2004a). Neuronal oscillations in cortical networks. *Science*, 304(5679), 1926–1929. <https://doi.org/10.1126/science.1099745>
- Buzsáki, G., & Draguhn, A. (2004b, June). Neuronal oscillations in cortical networks. <https://doi.org/10.1126/science.1099745>
- Buzsáki, G., Logothetis, N., & Singer, W. (2013). Scaling brain size, keeping timing: Evolutionary preservation of brain rhythms [Cross-species conservation of oscillatory architecture]. *Neuron*, 80(3), 751–764. <https://doi.org/10.1016/j.neuron.2013.10.002>
- Canolty, R. T., Edwards, E., Dalal, S. S., Soltani, M., Nagarajan, S. S., Kirsch, H. E., Berger, M. S., Barbaro, N. M., & Knight, R. T. (2006a). High gamma power is phase-locked to theta oscillations in human neocortex. *Science*, 313(5793), 1626–1628. <https://doi.org/10.1126/science.1128115>
- Canolty, R. T., Edwards, E., Dalal, S. S., Soltani, M., Nagarajan, S. S., Kirsch, H. E., Berger, M. S., Barbaro, N. M., & Knight, R. T. (2006b). High gamma power is phase-locked to theta oscillations in human neocortex. *Science*, 313(5793), 1626–1628. <https://doi.org/10.1126/science.1128115>
- Canolty, R. T., & Knight, R. T. (2010a). The functional role of cross-frequency coupling [Phase-amplitude coupling and multi-scale information transfer]. *Trends in Cognitive Sciences*, 14(11), 506–515. <https://doi.org/10.1016/j.tics.2010.09.001>
- Canolty, R. T., & Knight, R. T. (2010b). The functional role of cross-frequency coupling. *Trends in Cognitive Sciences*, 14(11), 506–515. <https://doi.org/10.1016/j.tics.2010.09.001>

- Cardin, J. A., Carlén, M., Meletis, K., Knoblich, U., Zhang, F., Deisseroth, K., Tsai, L.-H., & Moore, C. I. (2009). Driving fast-spiking cells induces gamma rhythm and controls sensory responses [Optogenetic evidence for PV interneuron gamma generation at 40 Hz]. *Nature*, *459*(7247), 663–667. <https://doi.org/10.1038/nature08002>
- Carter, G. C., Knapp, C. H., & Nuttall, A. H. (1973). Estimation of the magnitude-squared coherence function via overlapped fast Fourier transform processing. *IEEE Transactions on Audio and Electroacoustics*, *21*(4), 337–344. <https://doi.org/10.1109/TAU.1973.1162496>
- Cherry, N. (2002). Schumann resonances, a plausible biophysical mechanism for the human health effects of solar/geomagnetic activity. *Natural Hazards*, *26*(3), 279–331. <https://doi.org/10.1023/A:1015637127504>
- Cohen, M. X. (2017). Where does EEG come from and what does it mean? *Trends in Neurosciences*, *40*(4), 208–218. <https://doi.org/10.1016/j.tins.2017.02.004>
- Colgin, L. L., Denninger, T., Fyhn, M., Hafting, T., Bonnevie, T., Jensen, O., Moser, M.-B., & Moser, E. I. (2009). Frequency of gamma oscillations routes flow of information in the hippocampus [Slow vs fast gamma in hippocampal information routing]. *Nature*, *462*(7271), 353–357. <https://doi.org/10.1038/nature08573>
- Deco, G., & Kringelbach, M. L. (2017). Hierarchy of information processing in the brain: A novel ‘intrinsic ignition’ framework [Intrinsic ignition and metastable dynamics]. *Neuron*, *94*(5), 961–968. <https://doi.org/10.1016/j.neuron.2017.03.028>
- Dehaene, S., & Changeux, J.-P. (2011). Experimental and theoretical approaches to conscious processing. *Neuron*, *70*(2), 200–227. <https://doi.org/10.1016/j.neuron.2011.03.018>
- Donoghue, T., Haller, M., Peterson, E. J., Varma, P., Sebastian, P., Gao, R., Noto, T., Lara, A. H., Wallis, J. D., Knight, R. T., Shestiyuk, A., & Voytek, B. (2020). Parameterizing neural power spectra into periodic and aperiodic components. *Nature Neuroscience*, *23*(12), 1655–1665. <https://doi.org/10.1038/s41593-020-00744-x>
- Engel, A. K., & Fries, P. (2010). Beta-band oscillations—signalling the status quo? *Current Opinion in Neurobiology*, *20*(2), 156–165. <https://doi.org/10.1016/j.conb.2010.02.015>
- Faisal, A., Rahman, M. M., & Islam, M. R. (2023). EEG dataset for natural image recognition through visual stimuli. *Data in Brief*, *51*, 109761. <https://doi.org/10.1016/j.dib.2023.109761>
- Fernandez, L. M., & Lüthi, A. (2020). Sleep spindles: Mechanisms and functions. *Physiological Reviews*, *100*(2), 805–868. <https://doi.org/10.1152/physrev.00042.2018>
- Fries, P. (2015a). Rhythms for cognition: Communication through coherence. *Neuron*, *88*(1), 220–235. <https://doi.org/10.1016/j.neuron.2015.09.034>
- Fries, P. (2015b, October). Rhythms for cognition: Communication through coherence. <https://doi.org/10.1016/j.neuron.2015.09.034>
- Fries, P., Reynolds, J. H., Rorie, A. E., & Desimone, R. (2001). Modulation of oscillatory neuronal synchronization by selective visual attention. *Science*, *291*(5508), 1560–1563. <https://doi.org/10.1126/science.1055465>
- Friston, K. J., & Kiebel, S. (2009). Predictive coding under the free-energy principle [Hierarchical predictive coding and oscillatory dynamics]. *Philosophical Transactions of the Royal*

- Society B: Biological Sciences*, 364(1521), 1211–1221. <https://doi.org/10.1098/rstb.2008.0300>
- Gabor, D. (1946). Theory of communication. part 1: The analysis of information. *Journal of the Institution of Electrical Engineers-Part III: Radio and Communication Engineering*, 93(26), 429–441. <https://doi.org/10.1049/ji-3-2.1946.0074>
- Gaillard, R., Dehaene, S., Adam, C., Clémenceau, S., Hasboun, D., Baulac, M., Cohen, L., & Naccache, L. (2009). Converging intracranial markers of conscious access [Neural ignition and gamma signatures of conscious access]. *PLoS Biology*, 7(3), e1000061. <https://doi.org/10.1371/journal.pbio.1000061>
- Galambos, R., Makeig, S., & Talmachoff, P. J. (1981). A 40-hz auditory potential recorded from the human scalp [Discovery of 40 Hz ASSR resonance]. *Proceedings of the National Academy of Sciences*, 78(4), 2643–2647. <https://doi.org/10.1073/pnas.78.4.2643>
- Gramfort, A., Luessi, M., Larson, E., Engemann, D. A., Strohmeier, D., Brodbeck, C., Goj, R., Jas, M., Brooks, T., Parkkonen, L., & Hämäläinen, M. S. (2013). MEG and EEG data analysis with MNE-Python. *Frontiers in Neuroscience*, 7, 267. <https://doi.org/10.3389/fnins.2013.00267>
- Grandy, T. H., Werkle-Bergner, M., Chicherio, C., Schmiedek, F., Lövdén, M., & Lindenberger, U. (2013). Peak individual alpha frequency qualifies as a stable neurophysiological trait marker in healthy younger and older adults [IAF stability despite cognitive training]. *Psychophysiology*, 50(6), 570–582. <https://doi.org/10.1111/psyp.12043>
- Haller, M., Donoghue, T., Peterson, E. J., Varma, P., Sebastian, P., Gao, R., Noto, T., Knight, R. T., Shestiyuk, A., & Voytek, B. (2018). Parameterizing neural power spectra [Preprint; later published as Donoghue et al. 2020]. *bioRxiv*, 299859. <https://doi.org/10.1101/299859>
- Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., ... Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362. <https://doi.org/10.1038/s41586-020-2649-2>
- Herrmann, C. S., Rach, S., Neuling, T., & Strüber, D. (2013). Transcranial alternating current stimulation: A review of the underlying mechanisms and modulation of cognitive processes [Comprehensive tACS review - entrainment mechanisms]. *Frontiers in Human Neuroscience*, 7, 279. <https://doi.org/10.3389/fnhum.2013.00279>
- Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95. <https://doi.org/10.1109/MCSE.2007.55>
- Hyman, J. M., Zilli, E. A., Paley, A. M., & Hasselmo, M. E. (2005). Medial prefrontal cortex cells show dynamic modulation with the hippocampal theta rhythm dependent on behavior [Hippocampal-cortical theta coordination]. *Hippocampus*, 15(6), 739–749. <https://doi.org/10.1002/hipo.20106>
- Jasper, H. H. (1958). The ten-twenty electrode system of the International Federation. *Electroencephalography and Clinical Neurophysiology*, 10, 371–375.

- Jensen, O., & Colgin, L. L. (2007). Cross-frequency coupling between neuronal oscillations [Early review of cross-frequency coupling mechanisms]. *Trends in Cognitive Sciences*, 11(7), 267–269. <https://doi.org/10.1016/j.tics.2007.05.003>
- Kasten, F. H., Dowsett, J., & Herrmann, C. S. (2016). Sustained aftereffect of α -tACS lasts up to 70 min after stimulation [Long-lasting alpha entrainment effects]. *Frontiers in Human Neuroscience*, 10, 245. <https://doi.org/10.3389/fnhum.2016.00245>
- Kim, Y. C., & Powers, E. J. (1979). Digital bispectral analysis and its applications to nonlinear wave interactions. *IEEE Transactions on Plasma Science*, 7(2), 120–131. <https://doi.org/10.1109/TPS.1979.4317207>
- Klimesch, W. (1999a). EEG alpha and theta oscillations reflect cognitive and memory performance: A review and analysis. *Brain Research Reviews*, 29(2-3), 169–195. [https://doi.org/10.1016/S0165-0173\(98\)00056-3](https://doi.org/10.1016/S0165-0173(98)00056-3)
- Klimesch, W. (1999b, April). Eeg alpha and theta oscillations reflect cognitive and memory performance: A review and analysis. [https://doi.org/10.1016/s0165-0173\(98\)00056-3](https://doi.org/10.1016/s0165-0173(98)00056-3)
- König, H. L., Krueger, A. P., Lang, S., & Sönning, W. (1981). Biologic effects of environmental electromagnetism [Earlier findings reported in König, H.L. (1974). EEG-Veränderungen durch atmosphärische Felder niedriger Frequenz. *Naturwissenschaften*, 61, 422-423]. *Springer-Verlag*.
- Krigolson, O. E., Williams, C. C., Norton, A., Hassall, C. D., & Colino, F. L. (2017). Choosing MUSE: Validation of a low-cost, portable EEG system for ERP research [Validation of Muse headband for research]. *Frontiers in Neuroscience*, 11, 109. <https://doi.org/10.3389/fnins.2017.00109>
- Kuramoto, Y. (1984). Chemical oscillations, waves, and turbulence. *Springer-Verlag*. <https://doi.org/10.1007/978-3-642-69689-3>
- Kwon, J. S., O'Donnell, B. F., Wallenstein, G. V., Greene, R. W., Hirayasu, Y., Nestor, P. G., Hasselmo, M. E., Potts, G. F., Shenton, M. E., & McCarley, R. W. (1999). Gamma frequency-range abnormalities to auditory stimulation in schizophrenia [40 Hz ASSR deficit in schizophrenia]. *Archives of General Psychiatry*, 56(11), 1001–1005. <https://doi.org/10.1001/archpsyc.56.11.1001>
- Lachaux, J.-P., Rodriguez, E., Martinerie, J., & Varela, F. J. (1999). Measuring phase synchrony in brain signals. *Human Brain Mapping*, 8(4), 194–208. [https://doi.org/10.1002/\(SICI\)1097-0193\(1999\)8:4<194::AID-HBM4>3.0.CO;2-C](https://doi.org/10.1002/(SICI)1097-0193(1999)8:4<194::AID-HBM4>3.0.CO;2-C)
- Lisman, J., & Jensen, O. (2013, March). The theta-gamma neural code. <https://doi.org/10.1016/j.neuron.2013.03.007>
- Lisman, J. E., & Jensen, O. (2013). The theta-gamma neural code. *Neuron*, 77(6), 1002–1016. <https://doi.org/10.1016/j.neuron.2013.03.007>
- Little, S., & Brown, P. (2013). The functional role of beta oscillations in Parkinson's disease [Beta and akinesia in PD]. *Parkinsonism & Related Disorders*, 19(10), 943–950. <https://doi.org/10.1016/j.parkreldis.2013.07.010>
- Livio, M. (2002). *The golden ratio: The story of phi, the world's most astonishing number*. Broadway Books.

- Mashour, G. A., Roelfsema, P., Changeux, J.-P., & Dehaene, S. (2020). Conscious processing and the global neuronal workspace hypothesis [Updated review of GNW theory with neural signatures of conscious access]. *Neuron*, 105(5), 776–798. <https://doi.org/10.1016/j.neuron.2020.01.026>
- McKinney, W. (2010). Data structures for statistical computing in Python. *Proceedings of the 9th Python in Science Conference*, 56–61. <https://doi.org/10.25080/Majora-92bf1922-00a>
- Medendorp, W. P., van Deursen, K., Bijma, F., & Carandini, M. (2024). Neural oscillations in cephalopods show convergent evolution of sleep-related rhythms [12-18 Hz spindle-like oscillations in octopus - convergent evolution]. *iScience*, 27(3), 109145. <https://doi.org/10.1016/j.isci.2024.109145>
- Nickolaenko, A., & Hayakawa, M. (2014). *Schumann resonance for tyros: Essentials of global electromagnetic resonance in the Earth-ionosphere cavity*. Springer. <https://doi.org/10.1007/978-4-431-54358-9>
- Nikulin, V. V., Nolte, G., & Curio, G. (2011). A novel method for reliable and fast extraction of neuronal EEG/MEG oscillations on the basis of spatio-spectral decomposition. *NeuroImage*, 55(4), 1528–1535. <https://doi.org/10.1016/j.neuroimage.2011.01.057>
- Persinger, M. A. (2008). On the possible representation of the electromagnetic equivalents of all human memory within the Earth’s magnetic field: Implications for theoretical biology. *Theoretical Biology Insights*, 1, 3–11. <https://doi.org/10.4137/TBI.S726>
- Pobachenko, S. V., Kolesnik, A. G., Borodin, A. S., & Kalyuzhin, V. V. (2006). The contingency of parameters of human encephalograms and Schumann resonance electromagnetic fields revealed in monitoring studies. *Biophysics*, 51(3), 480–483. <https://doi.org/10.1134/S0006350906030225>
- Pogosyan, A., Gaynor, L. D., Eusebio, A., & Brown, P. (2009). Boosting cortical activity at beta-band frequencies slows movement in humans [First causal evidence for 20 Hz tACS motor effects]. *Current Biology*, 19(19), 1637–1641. <https://doi.org/10.1016/j.cub.2009.07.074>
- Posthuma, D., Neale, M. C., Boomsma, D. I., & De Geus, E. J. C. (2001). Are smarter brains running faster? Heritability of alpha peak frequency, IQ, and their interrelation [Genetic basis of IAF]. *Behavior Genetics*, 31(6), 567–579. <https://doi.org/10.1023/A:1013345411774>
- Price, C. (2016). Elf electromagnetic waves from lightning: The Schumann resonances. *Atmosphere*, 7(9), 116. <https://doi.org/10.3390/atmos7090116>
- Price, C., Williams, E., Elhalel, G., & Sentman, D. (2016). Natural ELF fields in the atmosphere and in living organisms [Critical review of SR-brain coupling hypotheses]. *International Journal of Biometeorology*, 60(7), 943–949. <https://doi.org/10.1007/s00484-015-1119-3>
- Saroka, K. S., Vares, D. E., & Persinger, M. A. (2016). Similar spectral power densities within the Schumann resonance and a large population of quantitative electroencephalographic profiles. *PLoS ONE*, 11(1), e0146595. <https://doi.org/10.1371/journal.pone.0146595>
- Sauseng, P., Klimesch, W., Heise, K. F., Gruber, W. R., Holz, E., Karim, A. A., Glennon, M., Gerloff, C., Birbaumer, N., & Hummel, F. C. (2009). Brain oscillatory substrates of visual short-term memory capacity [Theta-gamma coupling predicts working memory capacity]. *Current Biology*, 19(21), 1846–1852. <https://doi.org/10.1016/j.cub.2009.08.062>

- Schreiber, T. (2000). Measuring information transfer. *Physical Review Letters*, 85(2), 461–464. <https://doi.org/10.1103/PhysRevLett.85.461>
- Schumann, W. O. (1952). Über die strahlungslosen Eigenschwingungen einer leitenden Kugel, die von einer Luftschicht und einer Ionosphärenhülle umgeben ist. *Zeitschrift für Naturforschung A*, 7(2), 149–154. <https://doi.org/10.1515/zna-1952-0202>
- Siegel, M., Donner, T. H., & Engel, A. K. (2012). Spectral fingerprints of large-scale neuronal interactions. *Nature Reviews Neuroscience*, 13(2), 121–134. <https://doi.org/10.1038/nrn3137>
- Smit, D. J. A., Posthuma, D., Boomsma, D. I., & De Geus, E. J. C. (2006). Heritability of background EEG across the power spectrum [81% heritability of alpha frequency - genetic determination of frequency set points]. *Psychophysiology*, 43(6), 545–556. <https://doi.org/10.1111/j.1469-8986.2006.00448.x>
- Sohal, V. S., Zhang, F., Yizhar, O., & Deisseroth, K. (2009). Parvalbumin neurons and gamma rhythms enhance cortical circuit performance [PV interneurons and gamma oscillation function]. *Nature*, 459(7247), 698–702. <https://doi.org/10.1038/nature07991>
- Spitzer, B., & Haegens, S. (2017). Beyond the status quo: A role for beta oscillations in endogenous content (Re)activation [Extended beta function beyond status quo]. *eNeuro*, 4(4), ENEURO.0170–17.2017. <https://doi.org/10.1523/ENEURO.0170-17.2017>
- Strogatz, S. H. (2000). From Kuramoto to Crawford: Exploring the onset of synchronization in populations of coupled oscillators. *Physica D: Nonlinear Phenomena*, 143(1–4), 1–20. [https://doi.org/10.1016/S0167-2789\(00\)00094-4](https://doi.org/10.1016/S0167-2789(00)00094-4)
- Tognoli, E., & Kelso, J. A. S. (2014). The metastable brain [Metastable dynamics in large-scale brain networks]. *Neuron*, 81(1), 35–48. <https://doi.org/10.1016/j.neuron.2013.12.022>
- Tort, A. B. L., Komorowski, R., Eichenbaum, H., & Bhayasarathy, B. (2010). Measuring phase-amplitude coupling between neuronal oscillations of different frequencies. *Journal of Neurophysiology*, 104(2), 1195–1210. <https://doi.org/10.1152/jn.00106.2010>
- Uhlhaas, P. J., & Singer, W. (2010). Abnormal neural oscillations and synchrony in schizophrenia [Gamma oscillation deficits in schizophrenia]. *Nature Reviews Neuroscience*, 11(2), 100–113. <https://doi.org/10.1038/nrn2774>
- Varela, F., Lachaux, J.-P., Rodriguez, E., & Martinerie, J. (2001). The brainweb: Phase synchronization and large-scale integration. *Nature Reviews Neuroscience*, 2(4), 229–239. <https://doi.org/10.1038/35067550>
- Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., ... SciPy 1.0 Contributors. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17, 261–272. <https://doi.org/10.1038/s41592-019-0686-2>
- Wang, X.-J. (2010). Neurophysiological and computational principles of cortical rhythms in cognition [Comprehensive review including GABA-A time constants and oscillation frequencies]. *Physiological Reviews*, 90(3), 1195–1268. <https://doi.org/10.1152/physrev.00035.2008>

- Welch, P. D. (1967). The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms. *IEEE Transactions on Audio and Electroacoustics*, 15(2), 70–73. <https://doi.org/10.1109/TAU.1967.1161901>
- Zaehle, T., Rach, S., & Herrmann, C. S. (2010). Transcranial alternating current stimulation enhances individual alpha activity in human EEG [Alpha tACS entrainment aftereffects]. *PLoS One*, 5(11), e13766. <https://doi.org/10.1371/journal.pone.0013766>